



Commvault Professional Advanced

Student Guide

Module 3 – Storage Policies

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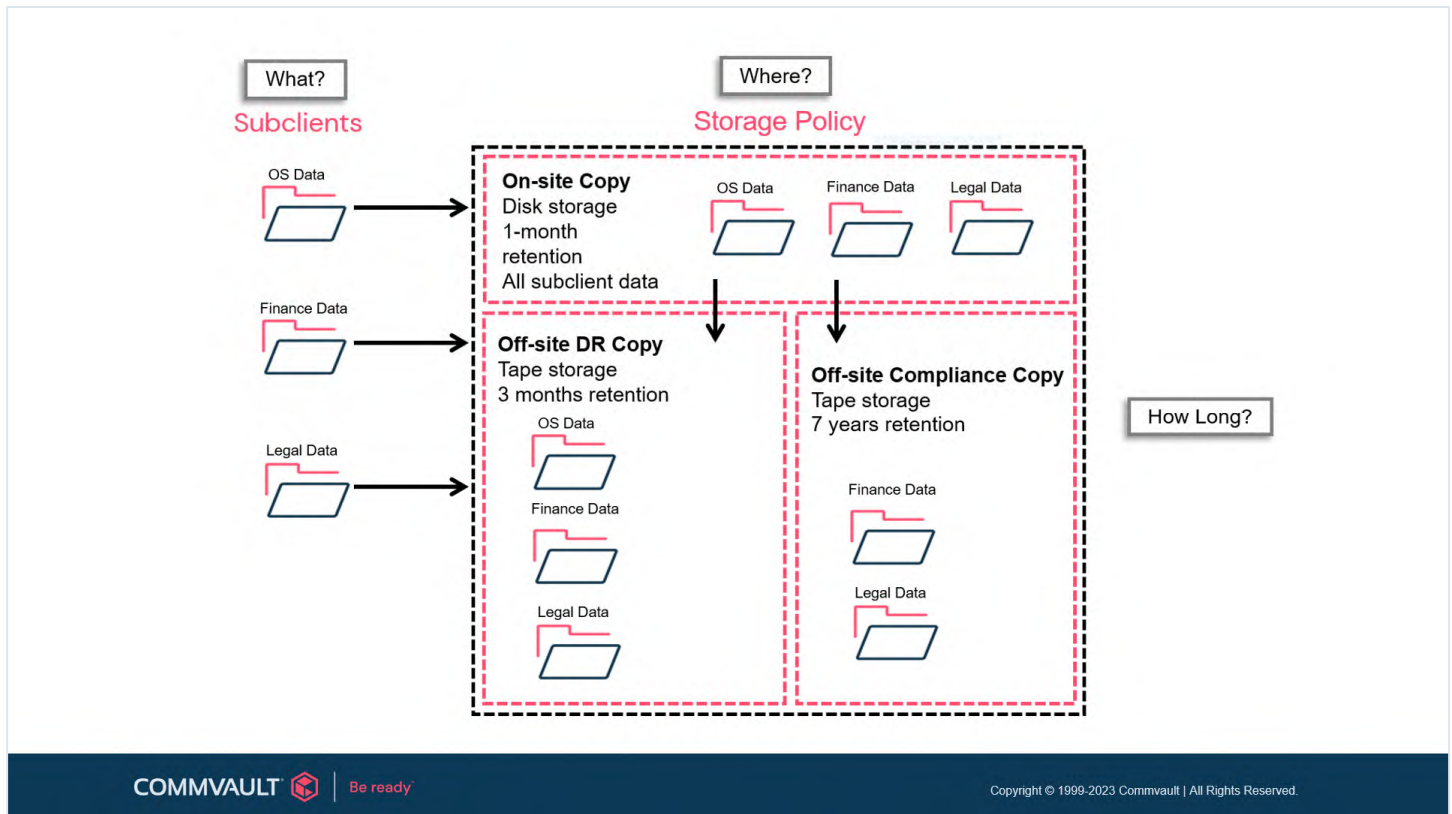
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Storage Policies



A storage policy defines the lifecycle management rules for all protected data, acting as a container with one or more policies that define how data will be managed. Within the storage policy, multiple copies can be created, each with their own set of rules.

Subclient (What Data)

For most file systems and applications, a default subclient is automatically generated. The default subclient protects all data for the agent. Additional subclients can be created to meet performance, management, and special protection requirements.

Data Path (Where data is stored)

The storage policy for the subclient determines the data path that moves data from the source location to protected storage. All active subclients must be associated with a storage policy. A data path includes a MediaAgent and library. If using a tape library, it also includes a drive pool and scratch pool.

Retention (How long data is kept)

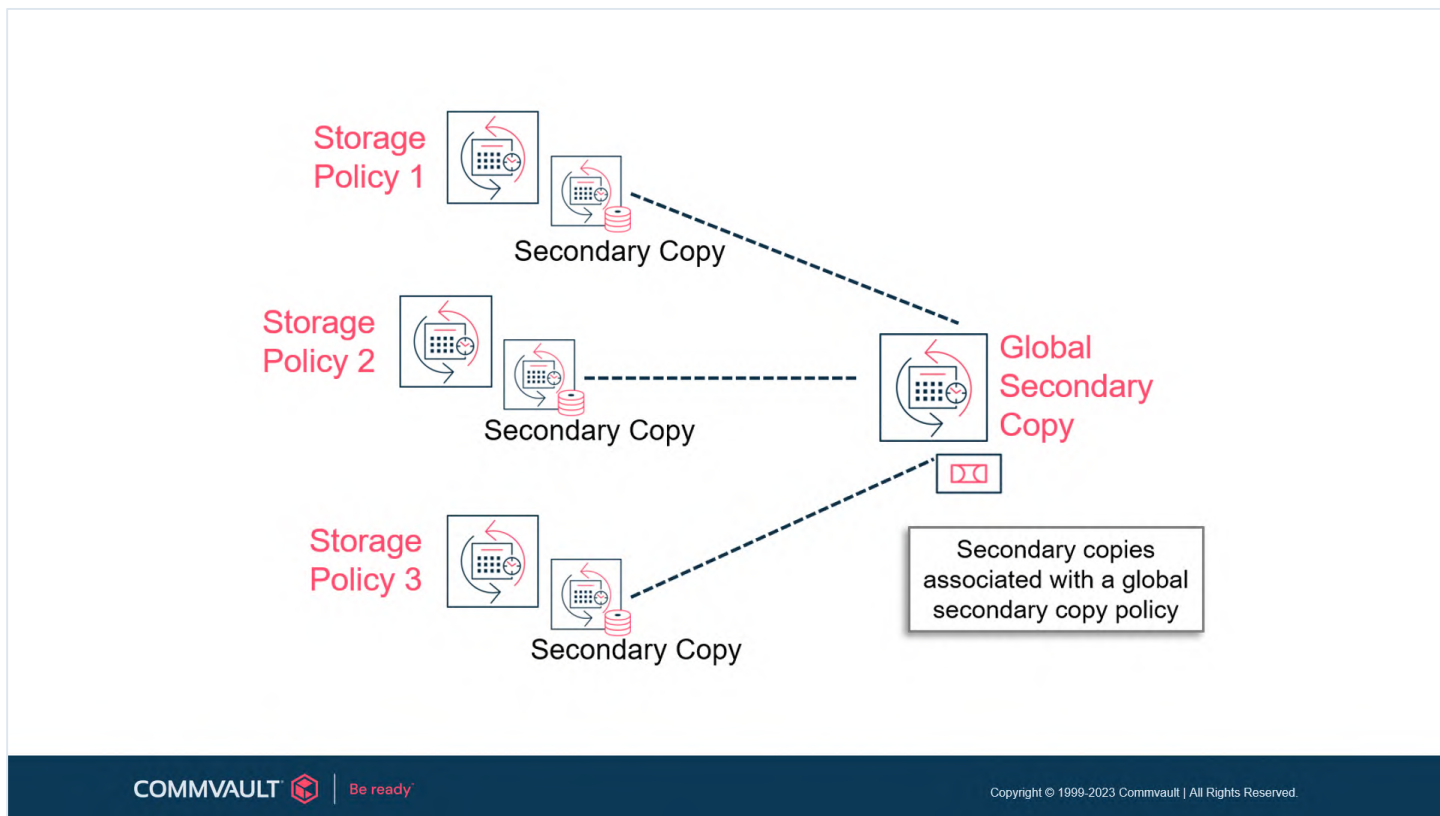
Each policy copy has its own retention configurations. This allows subclient data to be managed in multiple locations, each with their own retention settings.

Logical Divisions of Business Data

A storage policy creates logical boundaries for protected data. Data associated with and managed by a storage policy is bound to that policy. Protected data can be moved between copies within the same storage policy, but the data cannot be moved from one storage policy to another. This data isolation can be crucial when considering the management of data by different departments, different data types, different retention needs, or different storage locations.

Each storage policy has its own settings such as retention and copies. A storage policy also allows a single server to host different sets of data with different protection requirements by using multiple subclients.

Global Deduplication Policies

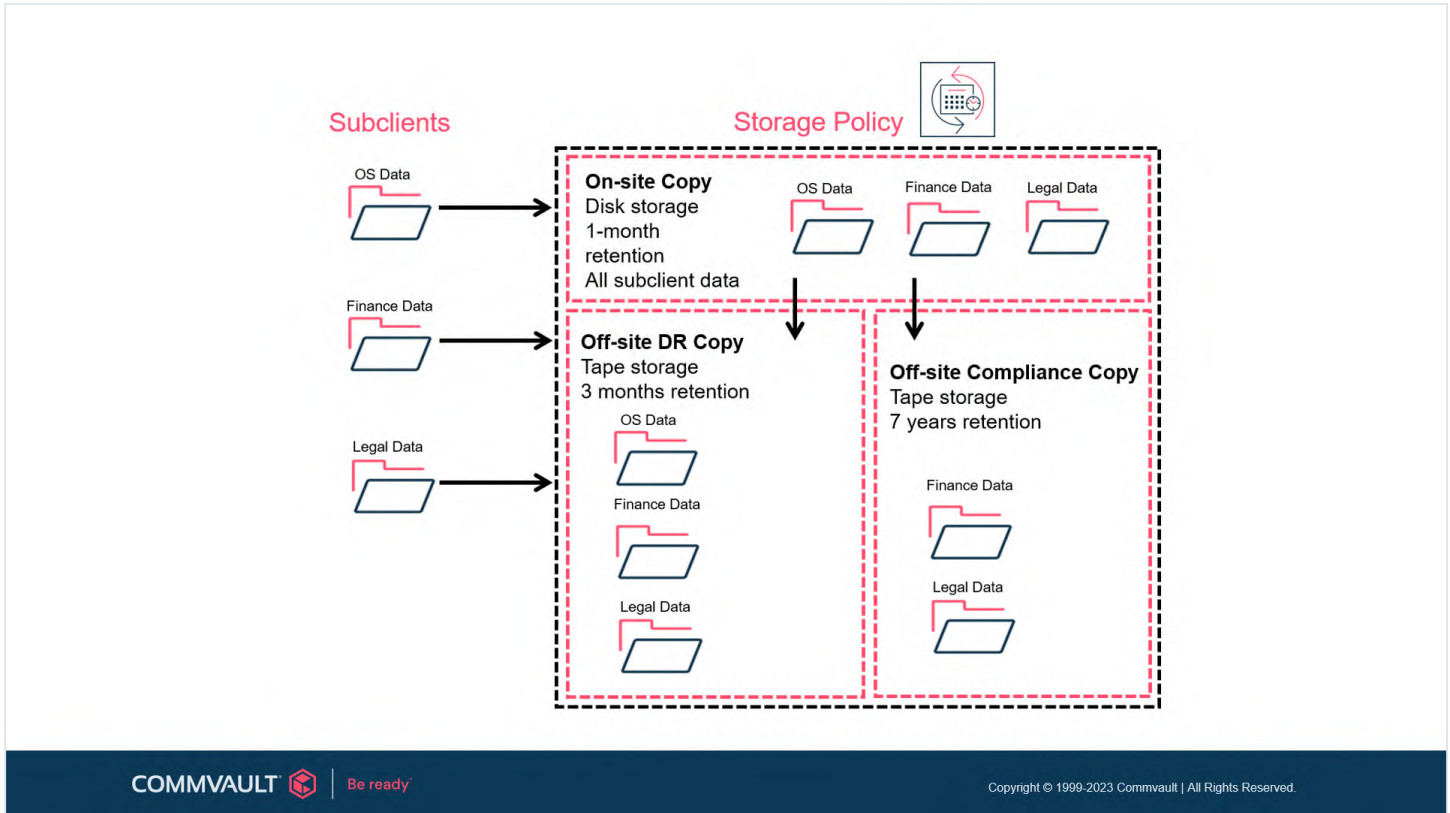


A global deduplication policy is not a storage policy, but it allows data from multiple storage policy copies to be combined into a single deduplicated disk storage structure. The global deduplication policy defines the location of the Deduplication Database (DDB) partitions and the disk library that contains the deduplication store.

Global Secondary Copies for Tape

A global secondary copy breaks the standard requirement that all data owned by a storage policy copy is placed in isolation on tape media. With a global secondary copy, you can assign data from multiple tape policy copies. This provides for more efficient tape media management, especially when consolidating data to tape from multiple remote sites, which may be configured with individual storage policies.

Storage Policy Structure



The concept of storage policy copies is that the data from the production environment only is moved to protected storage once. When data is in protected storage, the storage policy logically manages and maintains independent copies of that data. This allows for greater flexibility when managing data based on the three key aspects of data protection: disaster recovery, data recovery, and data archiving. There are five types of storage policy copies.

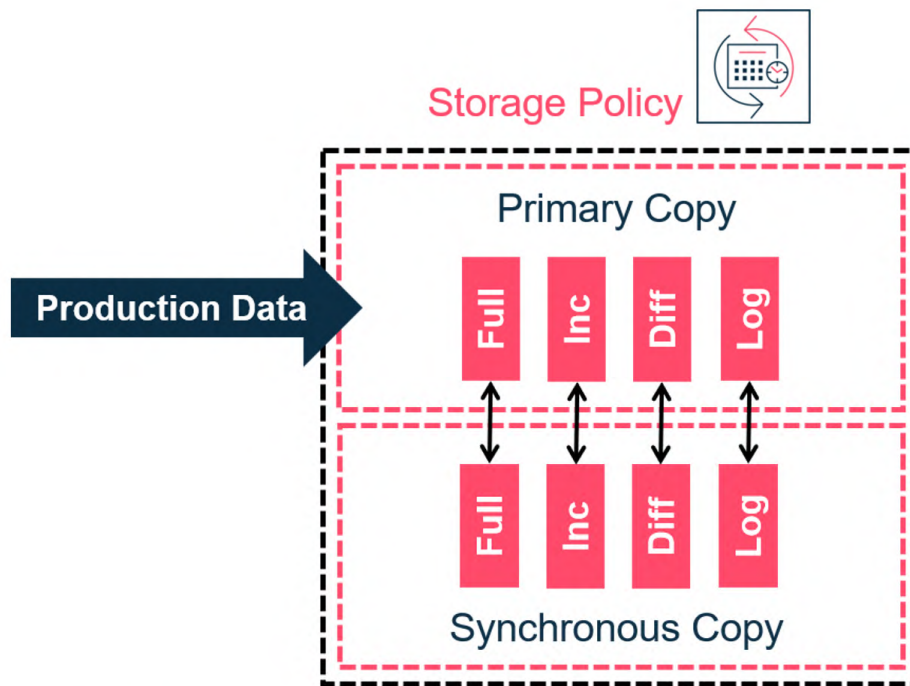
- Primary snap copy (used only with IntelliSnap[®] feature, block-level and VSA Application Aware backups)
- Primary backup copy
- Secondary synchronous copy
- Secondary selective copy
- Secondary snap copy

Primary Copy

A storage policy primary copy sets the primary rules for protected data. Each storage policy can have two primary copies:

- Primary snap copy – manages protected data using the Commvault IntelliSnap feature, any agents configured to run block-level backups or the virtual server agent (VSA) using application-aware backups.
- Primary backup copy (classic)– manages traditional agent-based data protection jobs. Most rules defined during the policy creation process are modified after it has been created.

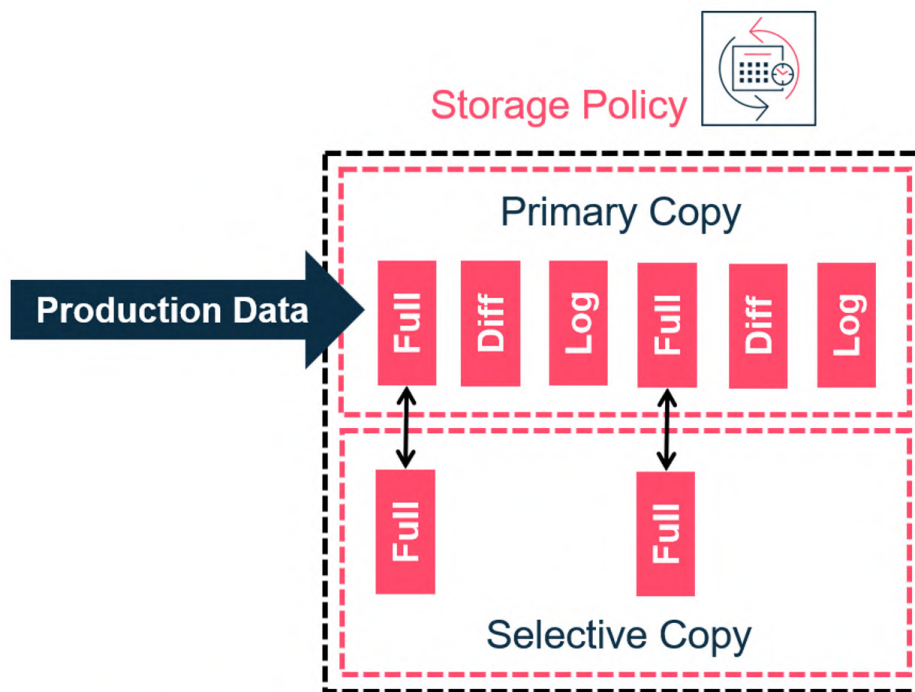
Secondary Synchronous Copy



Secondary Synchronous defines a secondary copy to synchronize protected data with a source copy. All valid data (jobs that completed successfully) written to the source copy are copied to the synchronous copy via an update process called an auxiliary copy operation. This means that all full, incremental, differential, transaction log, or archive jobs from a source copy are also managed by the synchronous copy. Synchronous copies are useful when you want a consistent point-in-time copy at any point within the cycle of all protected data available for restore. Synchronous copies are used to meet the following requirements:

- Provides consistent point-in-time copies of data required to restore data to a specific point-in-time within a cycle
- Provides copies that are required to be sent off-site daily
- Provides the ability to restore multiple versions of an object from a secondary copy within a cycle

Secondary Selective Copy



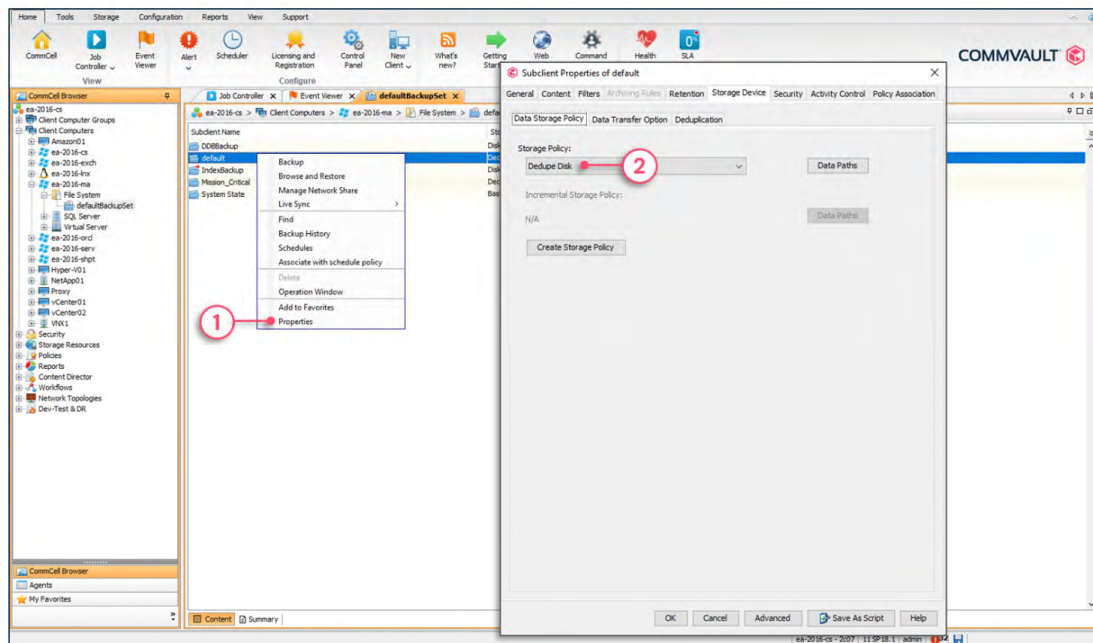
Secondary Selective Copy allows automatic selection of specific full backups or manual selection of any backup for additional protection. A selective copy can be scheduled on a specific cadence or frequency (number of cycles). Selective copies are used to meet the following requirements:

- Data being sent off-site weekly, monthly, quarterly, or yearly
- Archiving point-in-time copies of data for compliance and government regulations

Secondary Snap Copy

Secondary Snap Copy is used to manage clone, mirror, and vault copies. For NetApp arrays, multiple mirror and vault copies are created within a storage policy. For other vendors including EMC and HDS, an additional secondary snap copy is created to manage clone copies.

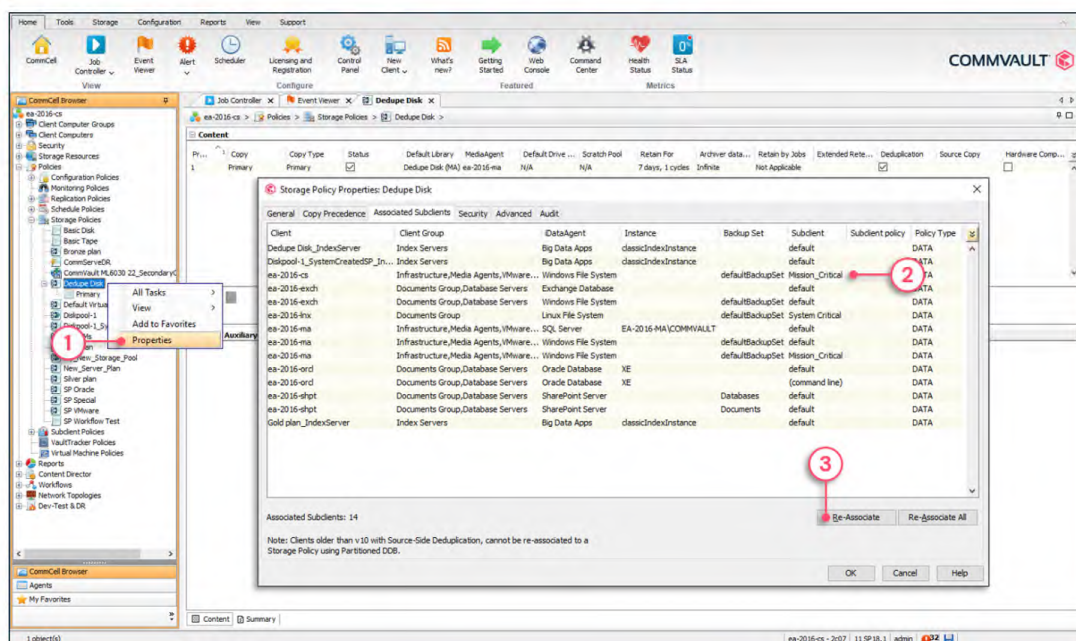
Subclient Association



To protect a subclient it must be associated with a storage policy. During an agent install, a storage policy is selected for the default subclient.

1. **Right-click the client > Properties**
2. Displays the Storage Policy used for this subclient

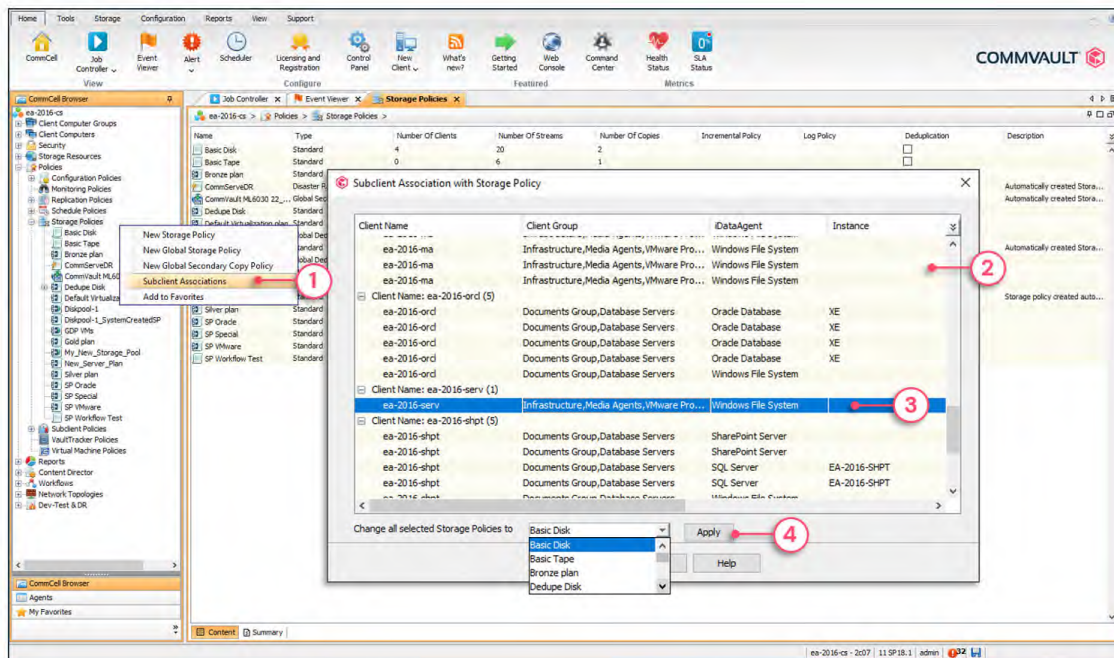
Storage Policy Level



All subclients for a specific storage policy are associated with another policy, but you can choose to associate them to a new policy.

1. **Expand Storage Policies > right-click the storage policy > Properties**
2. Displays all subclients associated to this storage policy
3. Click **Re-Associate** to re-associate the subclient to another storage policy

Subclient Association Policies

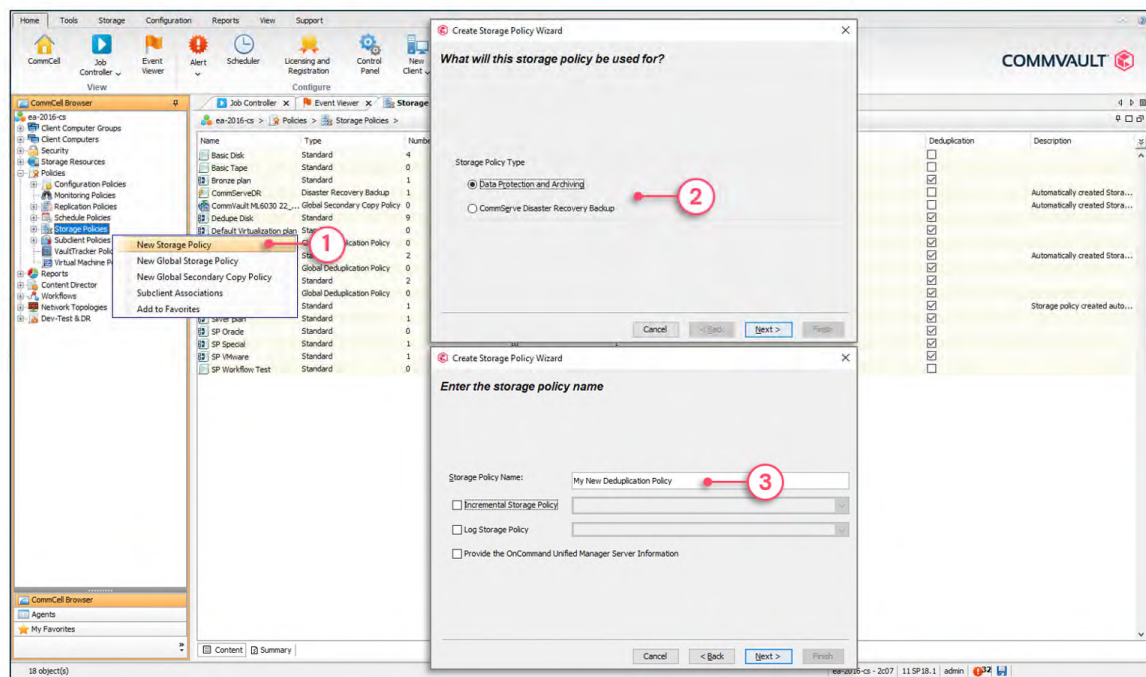


1. Expand Policies > right-click Storage Policies > Subclient Associations
2. Displays a list of all subclient-storage policy associations
3. Select the subclients to modify
4. Choose the storage policy from the dropdown list and click **Apply**

Storage Policy Wizard

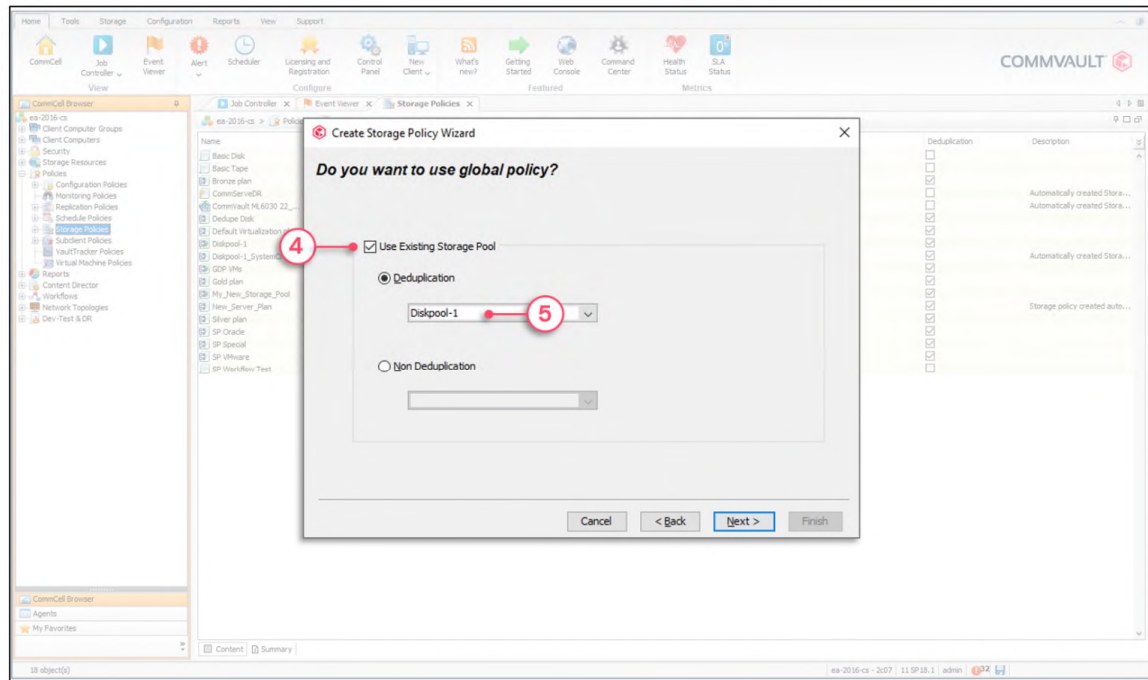
The Storage Policy Wizard is used to create storage policies for a CommCell environment. Storage policies are used for CommServe Disaster Recovery Backups or standard data protection. CommServe disaster recovery storage policies are only used for protecting the CommServe server metadata database, the CommServe registry, configuration files, and specified log files. No standard data can be protected by a CommServe DR policy. Standard data protection policies are used for protecting all production data within an environment.

Create a Storage Policy with Primary Copy Using a Global Deduplication Policy

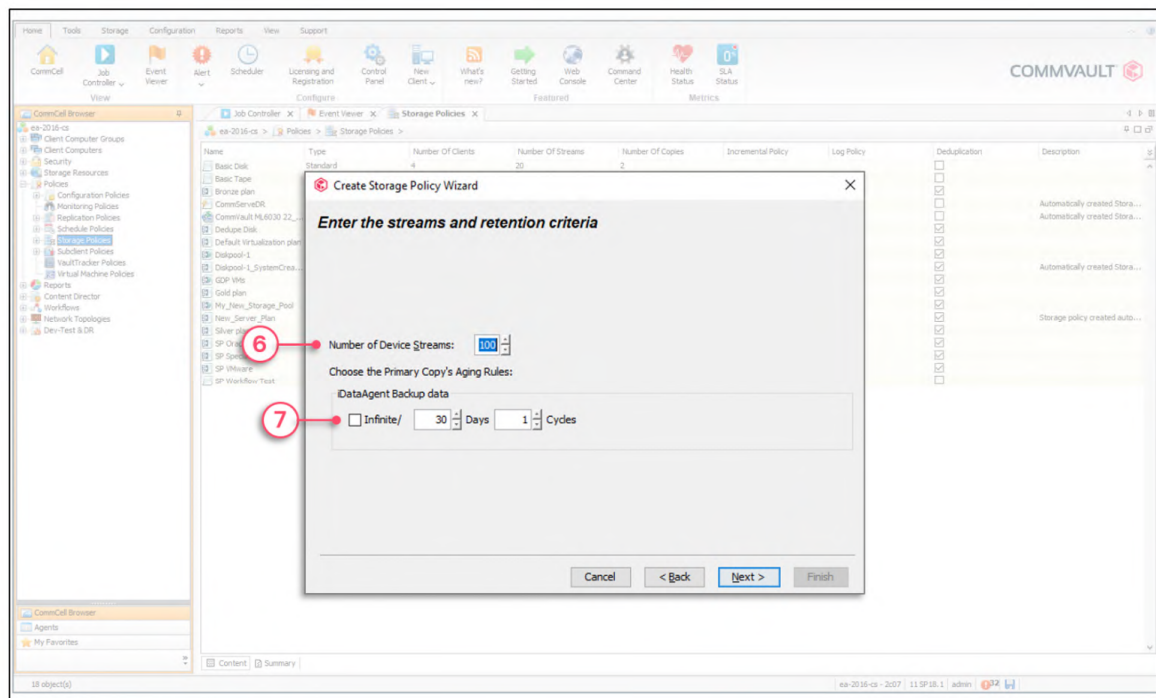


When the primary copy of the storage policy is using Commvault deduplication, it must be defined as a deduplicated storage policy during the creation process.

1. **Expand Policies > right-click Storage Policies > New Storage Policy**
2. Choose **Data Protection and Archiving** policy type
3. Enter a name

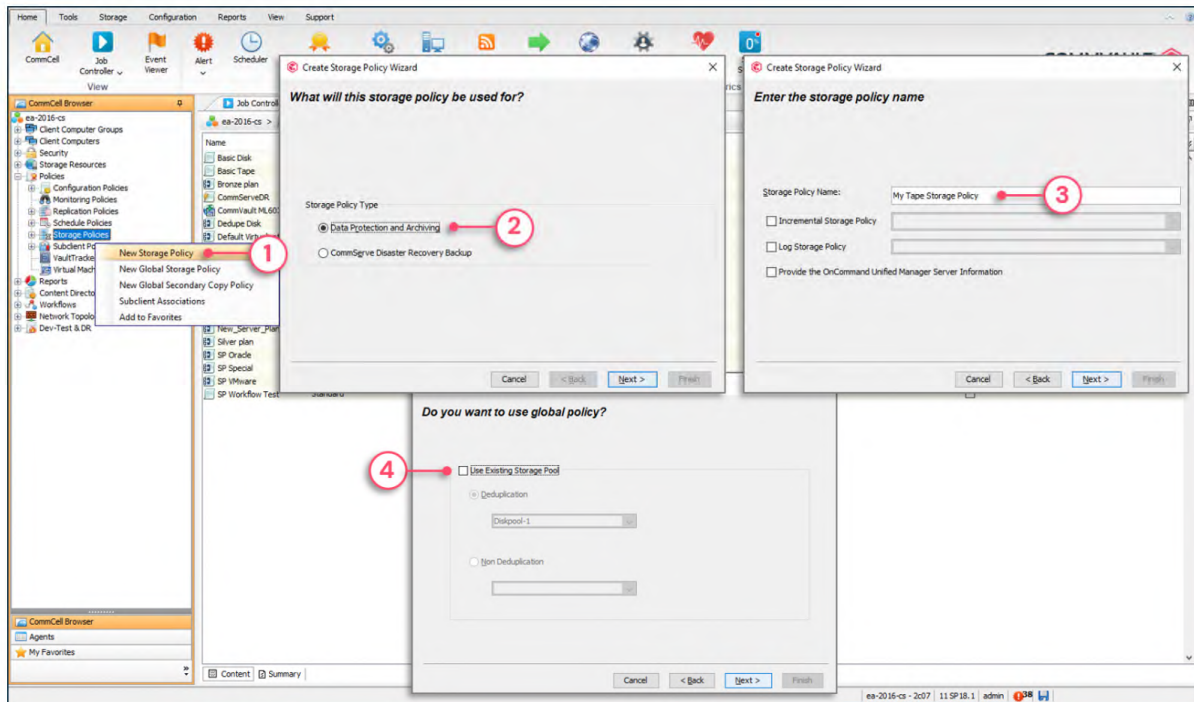


4. Check the box for **Use Existing Storage Pool** to use the existing storage pool
5. Select the Global Deduplication Policy



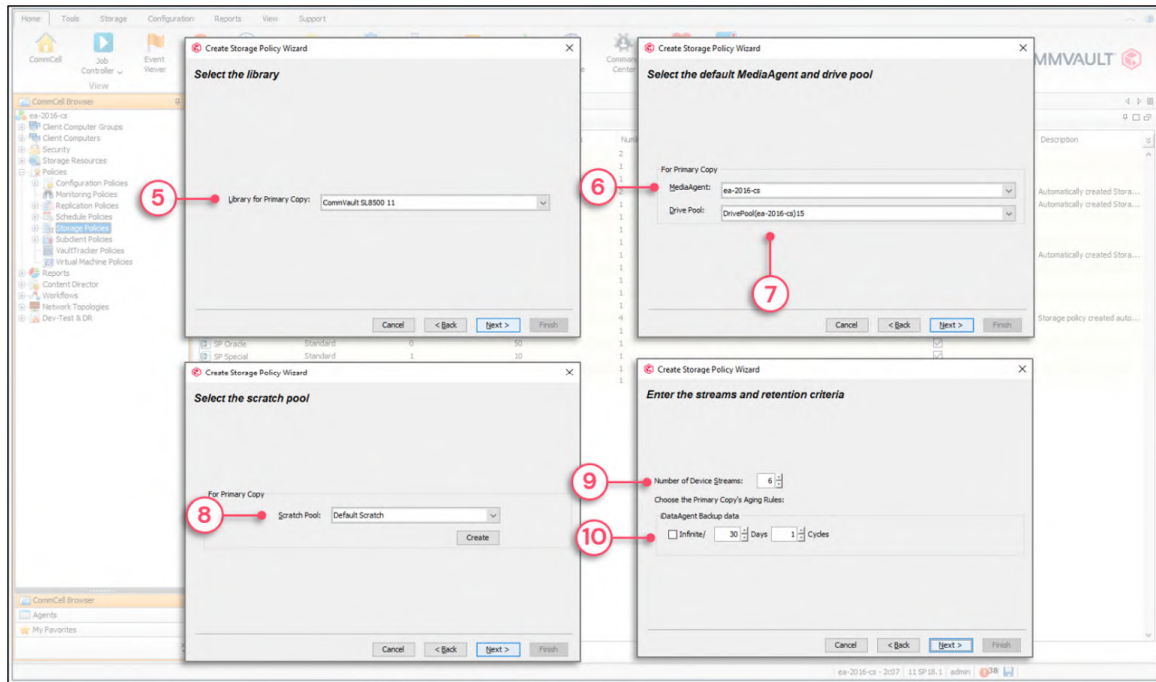
6. Define the number of streams
7. Set the retention for the data in the primary copy

Create a Storage Policy with Non-Deduplicated Primary Copy

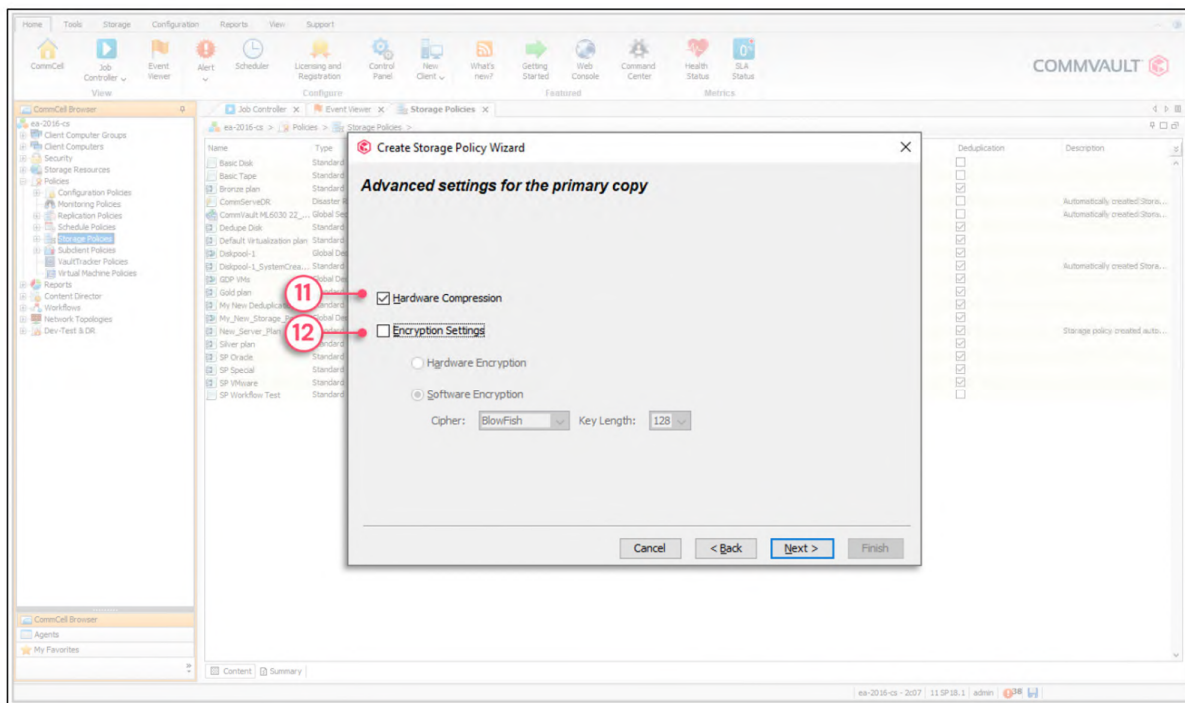


When the primary copy of the storage policy is not using Commvault deduplication, do not select the option to use a Global Deduplication Policy. Instead, specify the target library to use, such as tape.

1. **Expand Policies > right-click Storage Policies > New Storage Policy**
2. Select Storage Policy Type
3. Enter a name for the storage policy
4. Uncheck the box for **Use Existing Storage Pool** to create a non-deduplicated storage policy

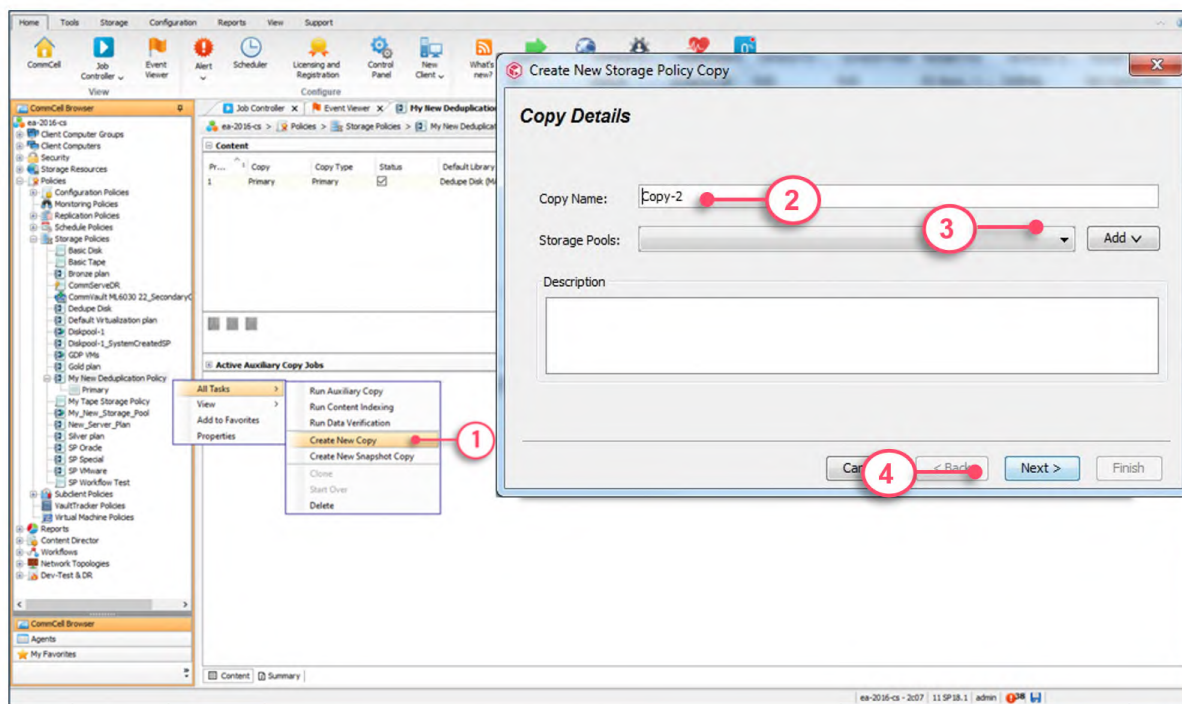


5. Select the library for the primary copy
6. Select the MediaAgent
7. For tape libraries, select the drive pool
8. Select the scratch pool
9. Define the number of streams
10. Define the retention for the data of the primary copy



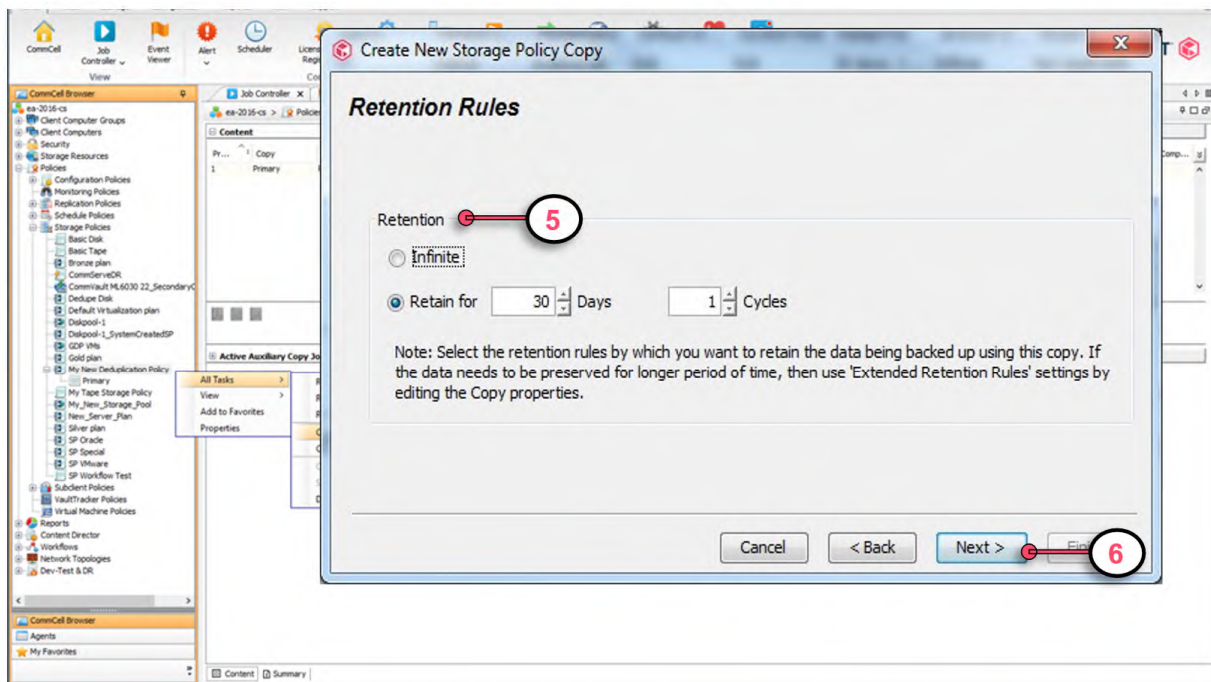
11. **Hardware Compression** is selected by default
12. Optionally, check the box for **Encryption Settings**

Secondary Copies

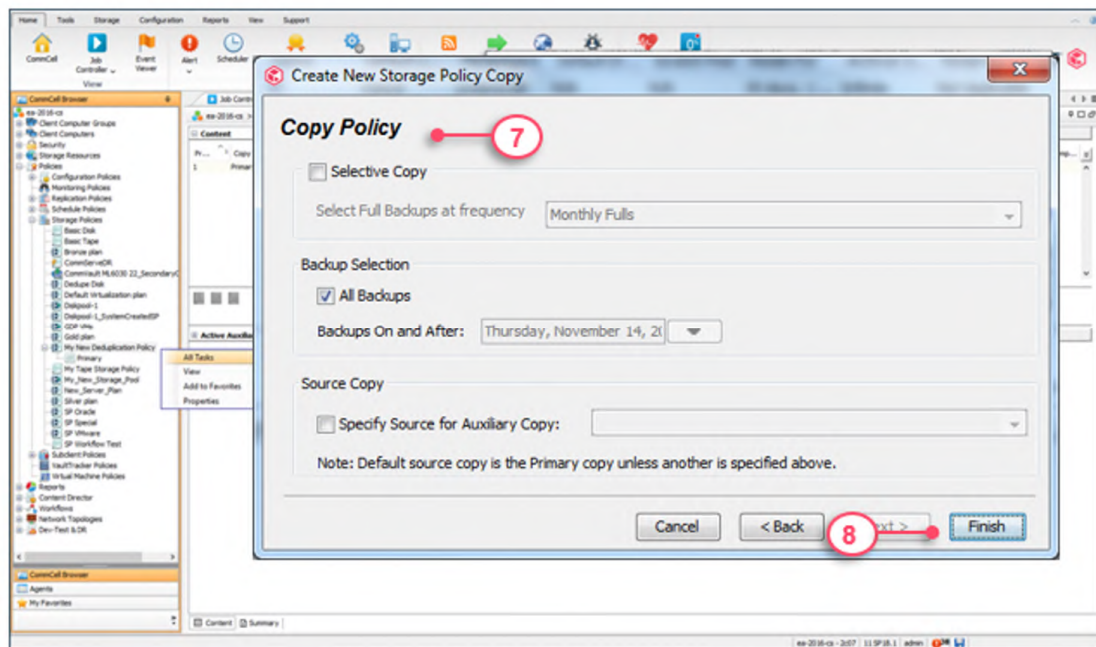


A storage policy can create more than one copy of the data. It is strongly recommended to create a secondary copy of the data to send offsite, such as on a set of tapes, to cloud or to a remote site disk library.

1. **Expand Storage Policies > right-click the storage policy > Create New Copy**
2. Name the storage policy copy
3. Define the tape library and MediaAgent that will be used for the copy
4. Click **Next**

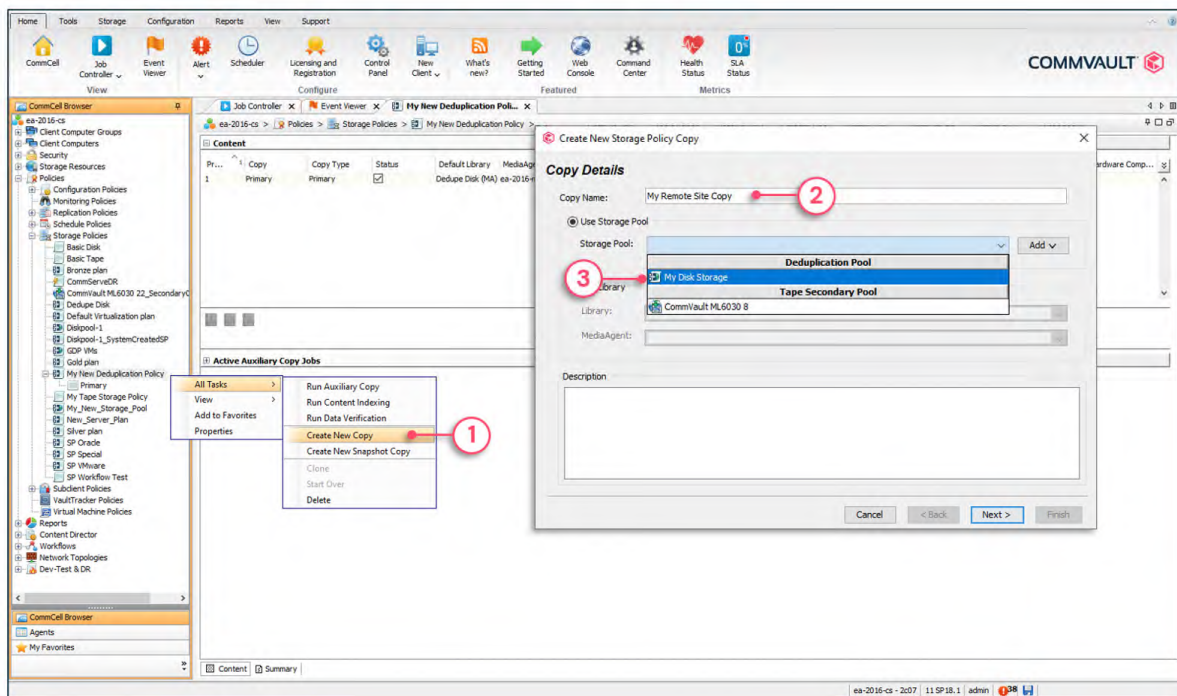


5. Select a retention option:
 - a. **Infinite:** If you plan to retain the data in the copy for an infinite amount of time
 - b. **Retain for n days n cycles:** If you plan to retain the data in the copy for a specific length of retention period and number of retention cycles. Retention cycles will not apply to agents that do not have cycles.
6. Click **Next**



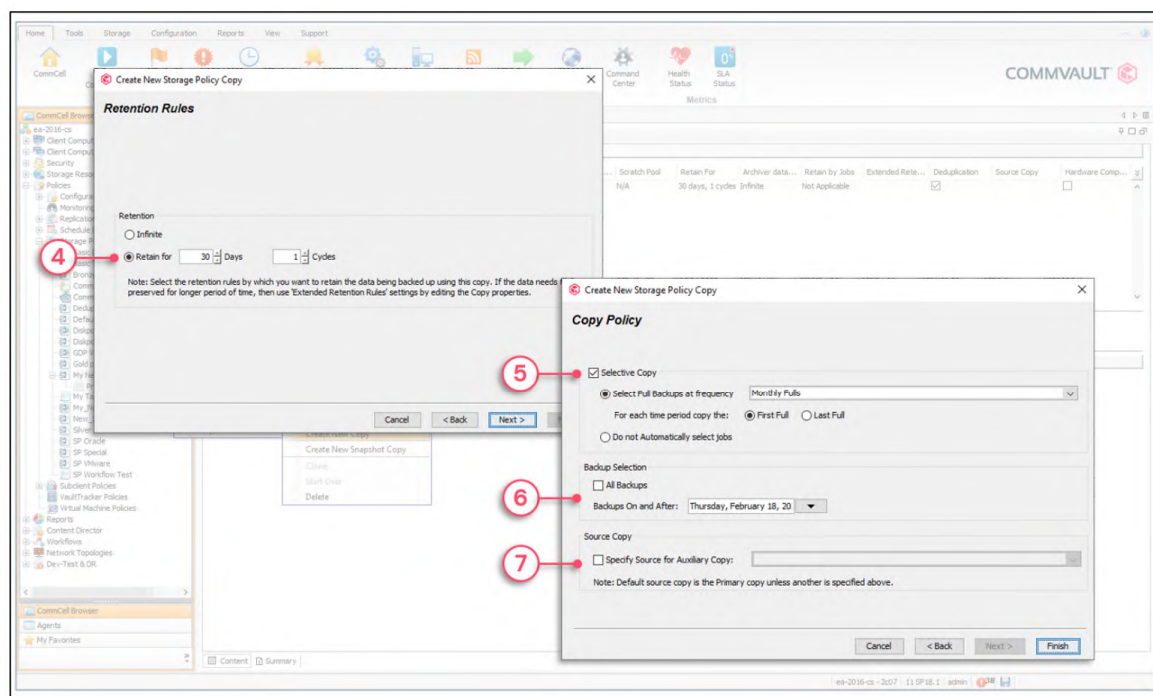
7. Define which copy type is used by choosing one or more copy policy options:
 - a. **All Backups:** Copies all the backup data in the primary copy to the secondary copy. If you do not use this option, the **Backups on and After** option is enabled.
 - b. **Backups on and After:** Copies only the backup data after the specified date from the primary copy to the secondary copy
8. Click **Finish**

Configure a Secondary Copy to Use a Global Deduplication Policy



If the secondary copy is sent to the cloud or to a remote site disk library, it can be deduplicated. The global deduplication policy must first be created. Then, create the secondary copy and select the global deduplication policy from the list.

1. **Expand Storage Policies > right-click the storage policy > Create New Copy**
2. Name to the storage policy copy
3. Select the remote site storage pool



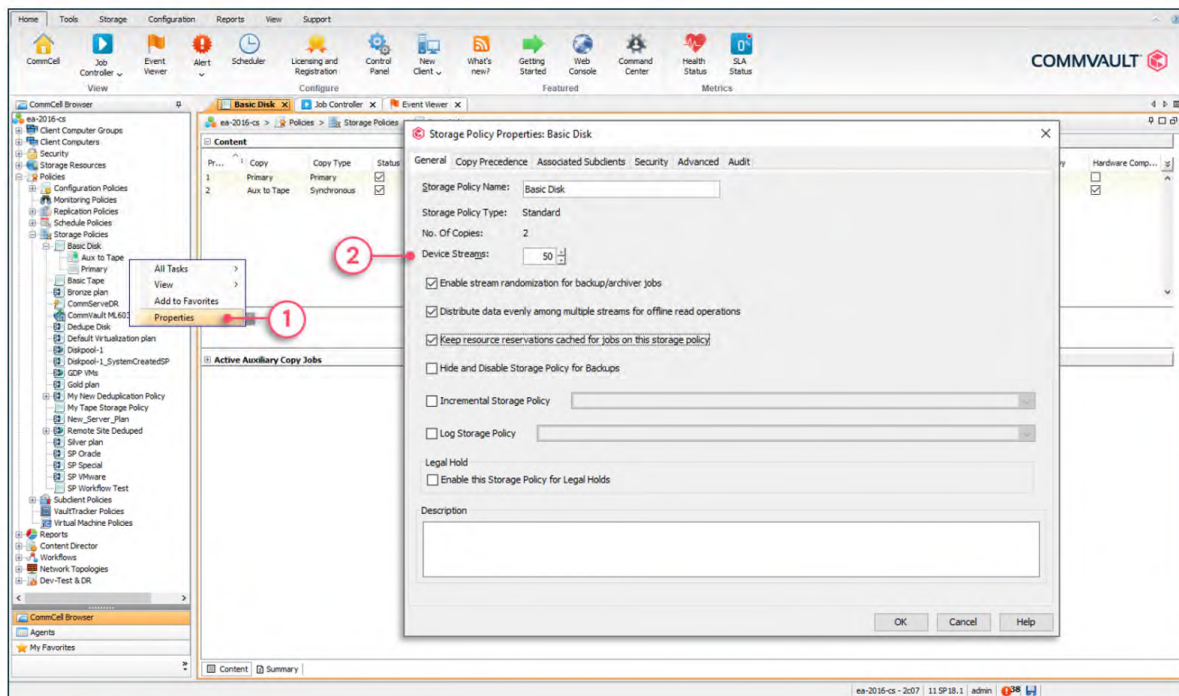
4. Define the retention for the backup data
5. Check the box for **Selective Copy**
6. Check the box for **All Backups** to copy all primary copies, or enter a date
7. Check the box for **Specify Source for Auxiliary Copy** if needed

Storage Policy Streams

Device streams determine how many concurrent write operations are performed to the library. Increase device streams to allow for more concurrent job streams to write if adequate resources are available.

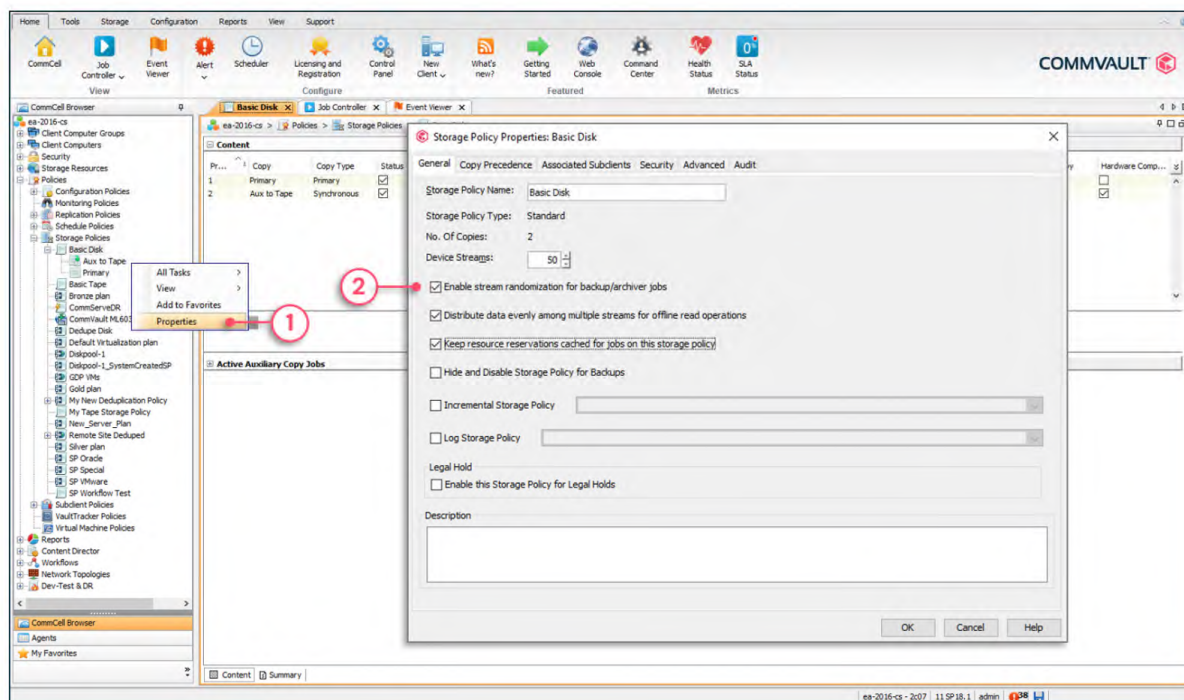
- MediaAgent – Ensure the MediaAgent is properly scaled to accommodate higher stream concurrency
- Network – Ensure the network bandwidth can manage higher traffic
- Disk Library without Commvault deduplication – Ensure the library can handle higher number of write operations; increase the number of mount path writers so the total number of writers across all mount paths equals the number of device streams
- Disk Library with Commvault deduplication – Enable client-side deduplication. Each DDB can manage 50 or more concurrent streams; if using client-side deduplication, after the initial full backup is complete most data processing is done locally on each client. This means minimum bandwidth, MediaAgent, and disk storage resources are required for data protection operations.
- Tape Library – Enable multiplexing if tape write speeds are slow

Configure Storage Policy Streams



1. Expand Storage Policies > right-click the storage policy > Properties
2. Define the number of streams for the storage policy

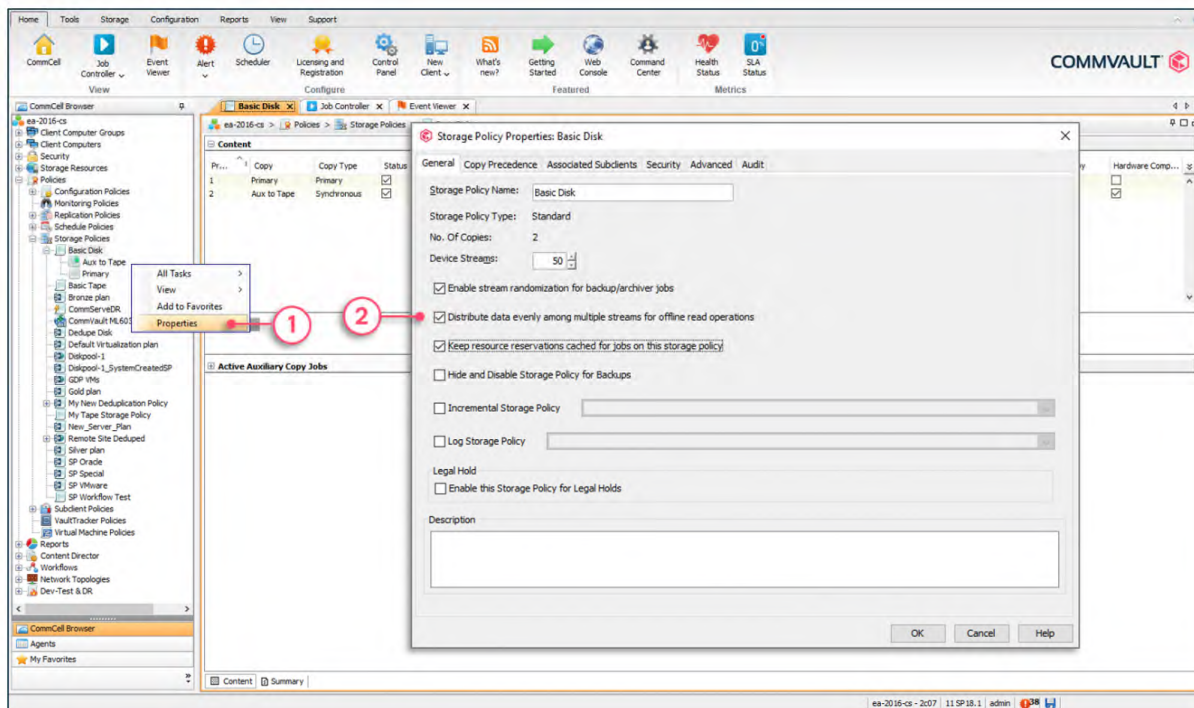
Stream Randomization



When conducting auxiliary copy jobs, stream randomization improves performance by randomly selecting source streams during the auxiliary copy job. This results in a more balanced stream of data when multi-streaming auxiliary copy jobs.

1. **Expand Storage Policies > right-click the storage policy > Properties**
2. Check the box to enable randomization

Distribute Data Evenly for Offline Read Operations

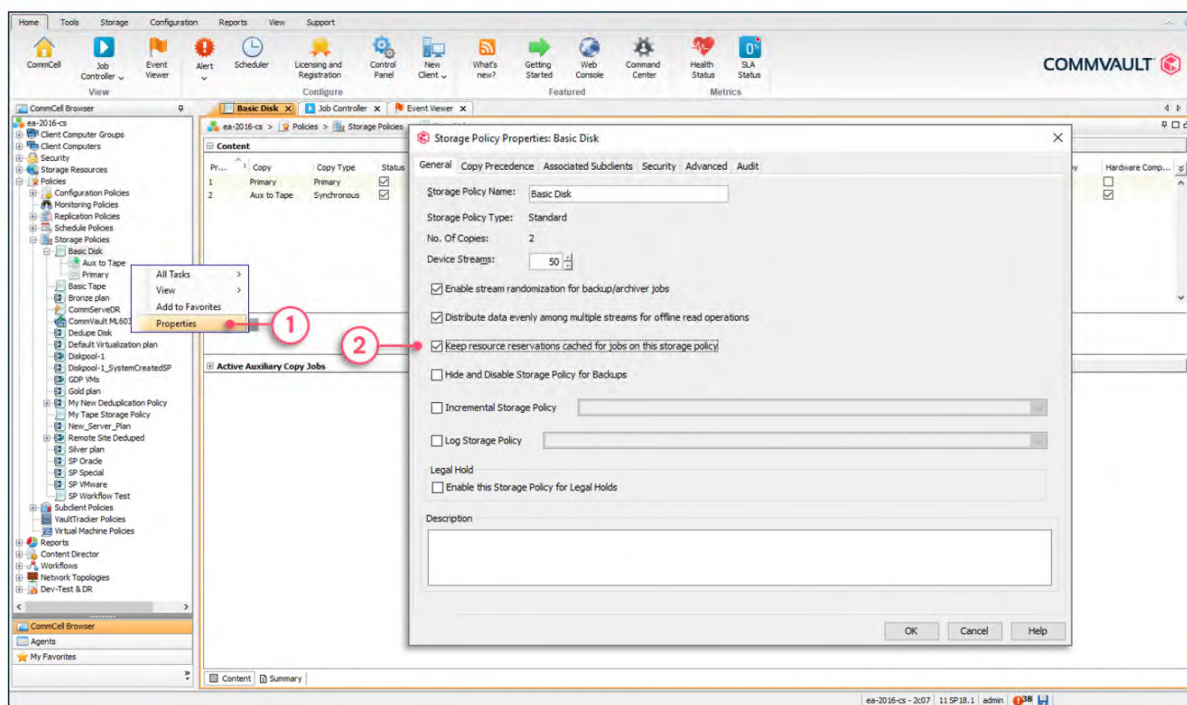


For auxiliary copy-based jobs, including content indexing and data verification, data is distributed across multiple job streams for better performance. This means that the system breaks incoming large streams from the client into smaller streams in the primary copy to facilitate the auxiliary copy of those streams.

For example, a large Exchange database that generates a 4TB stream while running an auxiliary copy of that single stream can take a significant amount of time. But breaking it down into smaller streams while receiving it helps the subsequent auxiliary process by leveraging multi-streaming. This results in copying more than one stream concurrently.

1. **Expand Storage Policies > right-click the storage policy > Properties**
2. Check the box for **Distribute data evenly...** to enable even distribution of streams

Cached Resource Reservations

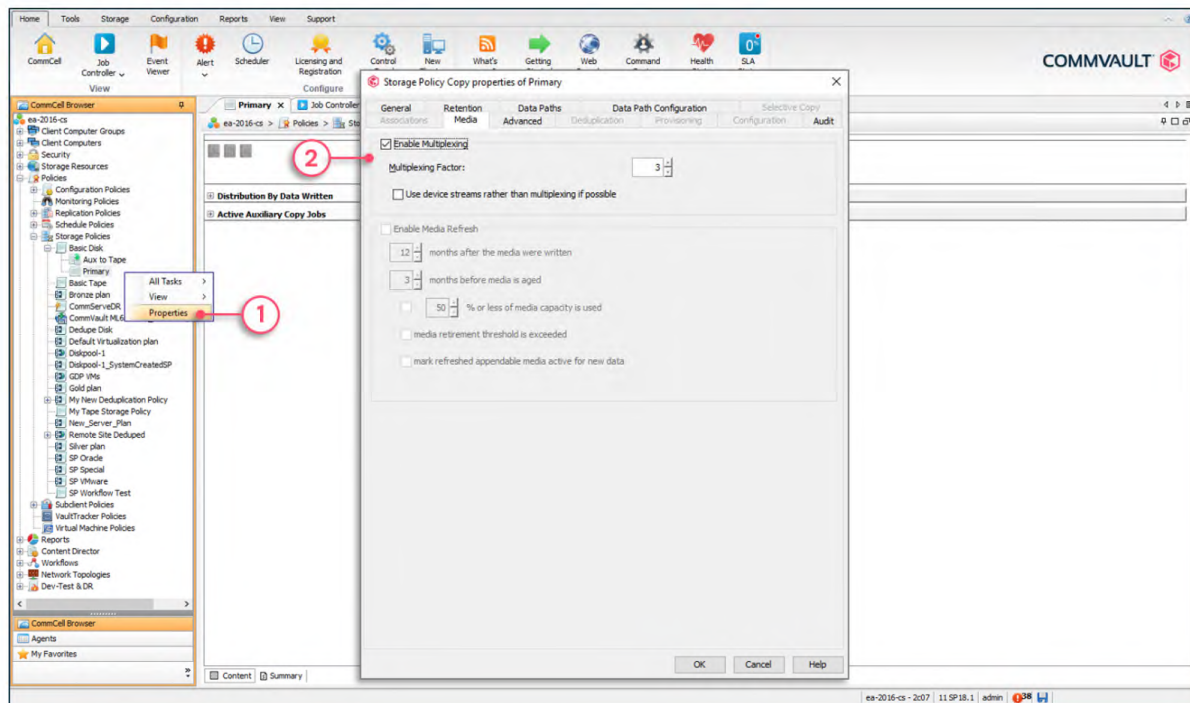


For large application servers conducting frequent transaction log backups, the 'keep resource reservations cached for jobs on this storage policy' option improves performance. Checking this option eliminates the step for the CommServe server to contact the MediaAgent to reserve storage resources each time a job is run.

Use this feature when the client is writing to a dedicated library where all resources are exclusive to the client, such as in client/MediaAgent LAN-Free backups.

1. **Expand Storage Policies > right-click the storage policy > Properties**
2. Check the box for **Keep resource reservations cached...** to reserve resources for this storage policy

Multiplexing



Multiplexing should only be used for tape storage. When writing multiple streams to a tape library, multiplexing is used to improve write performance by multiplexing multiple job streams into a device stream. Multiplexing improves backup performance but can have a negative effect on restore performance. If the source location is a disk library with multiple mount paths, this option can improve read performance from the disks when using the 'Combine to Streams' option.

When writing the primary copy to a disk library, there are no advantages to enabling multiplexing because the disk library already receives multiple streams concurrently from subclients, and if available, leverages multiple mount paths.

1. **Expand Storage Policies > right-click the Primary copy > Properties**
2. Enable multiplexing and set a multiplexing factor

Combine to Streams

A storage policy is configured to allow the use of multiple streams for primary copy backup. Multi-streaming of backup data is done to improve backup performance. Normally, each stream used for the primary copy requires a corresponding stream on each secondary copy.

If tape media is used for a secondary copy, multi-stream storage policies consume multiple media. The 'Combine to streams' option is used to consolidate multiple streams from source data on to fewer media when secondary copies are run. This allows for better media management and the grouping of like data onto media for storage.

Using 'Combine to streams' for Tape Media

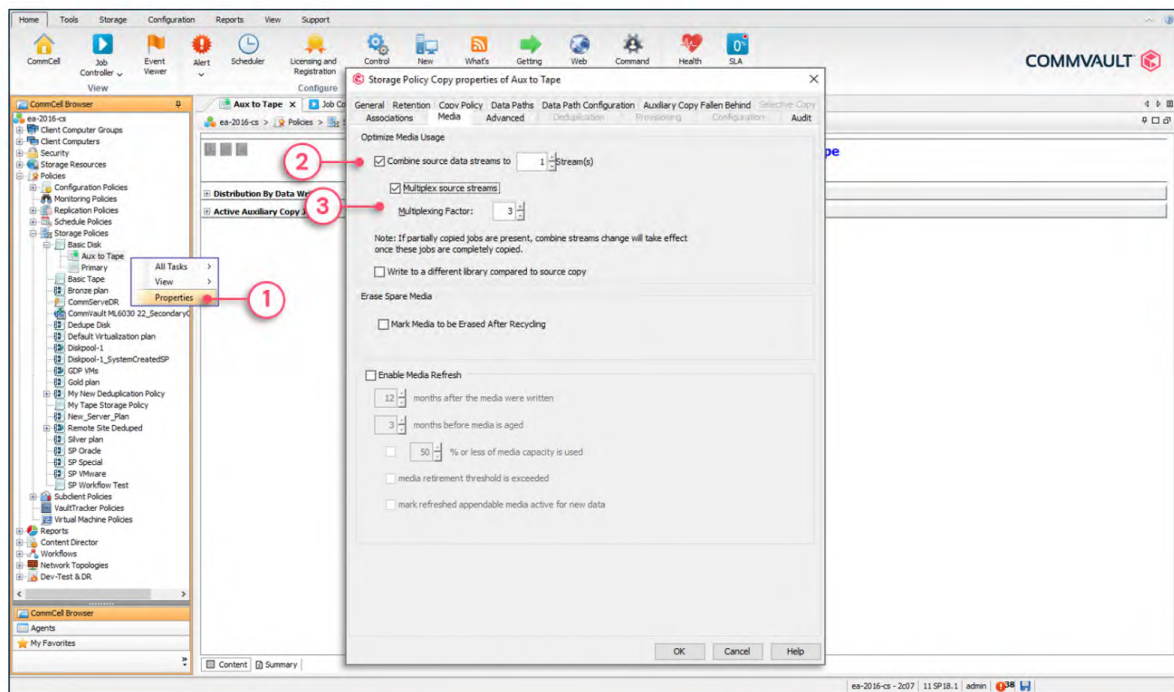
You back up home folders subclient to a disk library using three streams to maximize performance. The total size of protected data is 600GB. You want to consolidate those three streams onto a single 800GB capacity tape for off-site storage.

Solution: By creating a secondary copy and setting the 'Combine to streams' setting to 1 you will serially place each stream onto the media.

In some cases, using the 'Combine to streams' option may not be the best method to manage data. Multi-streaming backup data is done to improve performance. When those streams are consolidated to the same media set, they can only be recovered in a single stream operation. Though combining to streams has a media consolidation benefit, it will have a negative effect on the restore performance.

Another reason not to use the 'Combine to streams' option is for multi-streamed backups of SQL, DB2, and Sybase subclients. When these agents use a single subclient with multi-streaming enabled, the streams must be restored in the same sequence they were backed up in. If the streams are combined to the same tape, they must be pre-staged to disk before they can be recovered. In this case, not enabling 'Combine to streams' and placing each stream on separate media bypasses the pre-staging of the data and allows multiple streams to be restored concurrently, making the restore process considerably faster. Note that this only applies to subclients that have been multi-streamed. If multiple subclients have been single-streamed and combined to media, they do not need to be pre-staged prior to recovery.

Configure Storage Policy Secondary Copy Streams



1. Expand Storage Policies > right-click the storage policy secondary tape copy > Properties
2. Define the number of device streams, or tape drives, source streams are combined to
3. Define the number of streams that are multiplexed to each drive

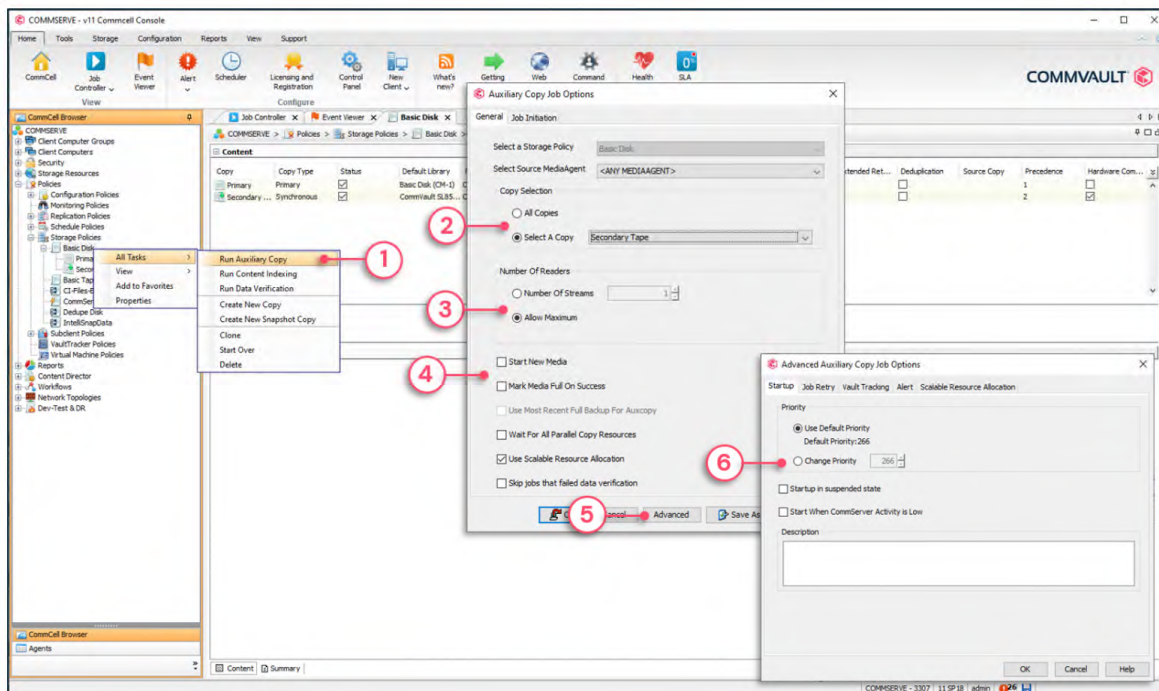
Auxiliary Copy Configuration

Auxiliary copy operation allows you to schedule, run on-demand, save a job as a script, or set an automatic copy. An auxiliary copy job is used to copy jobs to secondary copy locations, generate content indices, and to fully reconstruct the DDB.

With auxiliary copies you can:

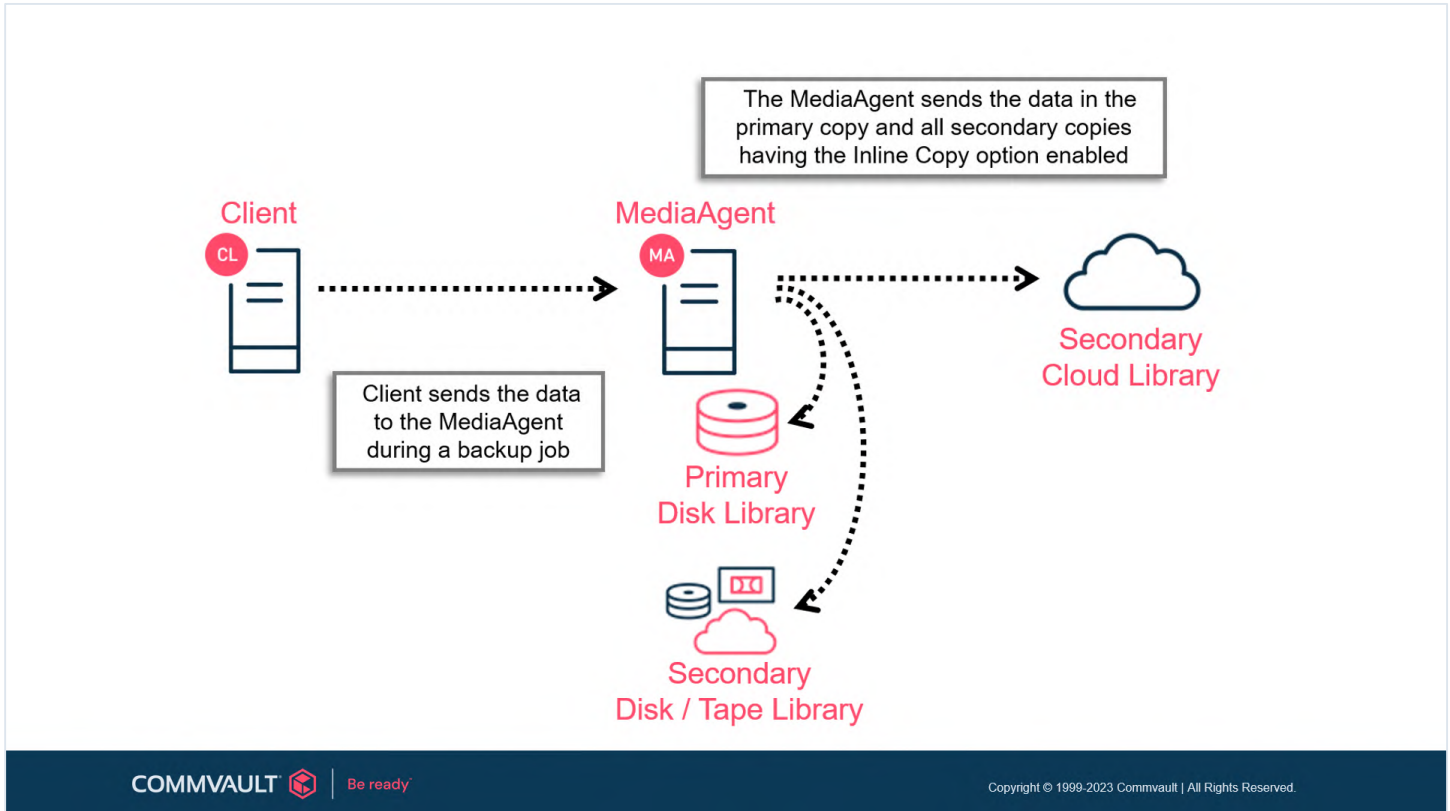
- Allocate number of readers or streams to use during auxiliary copy job
- Decide which secondary copies you want to include in the auxiliary copy job
- Start new media and mark media full, which can be used to isolate jobs on media
- Use VaultTracker options to export and track media using VaultTracker policies and reports
- Use job priorities to assign different job priorities for auxiliary copies

Configure Auxiliary Copy Options



1. Right-click the storage policy > All Tasks > Run Auxiliary Copy
2. Choose to run the auxiliary copy for all copies or choose a specific copy
3. Use the maximum or define a specific number of streams
4. Check the boxes for **Start New Media** and **Mark Media Full On Success** to isolate the job on a set of media
5. Click **Advanced** to open the advanced auxiliary copy job options
6. If needed, change the priority number of the job

Inline Copy



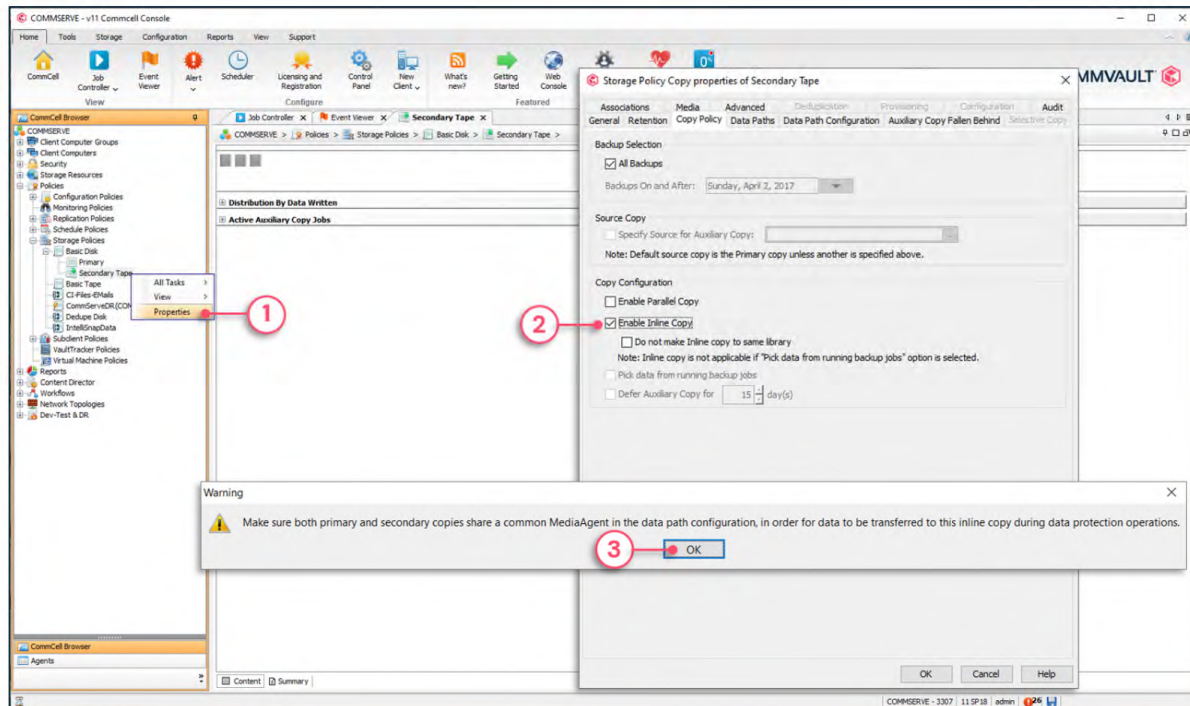
The inline copy option lets you create additional copies of data at the same time you are performing primary backups. This feature is useful when two copies of data are needed quickly. Data is passed from the client to the MediaAgent as job streams. The MediaAgent then creates two sets of device streams and sends each to the appropriate library. Although this is a quick method for creating multiple copies, there are a few caveats to consider.

- Inline Copy is not supported if client-side deduplication has been enabled
- If the primary copy fails, the secondary copy also fails
- Both copies are made at the same time, thus twice as many library resources are required; this may prevent other jobs from running and can cause overall performance to degrade, as the entire job will run at the pace of the slowest resource

Example:

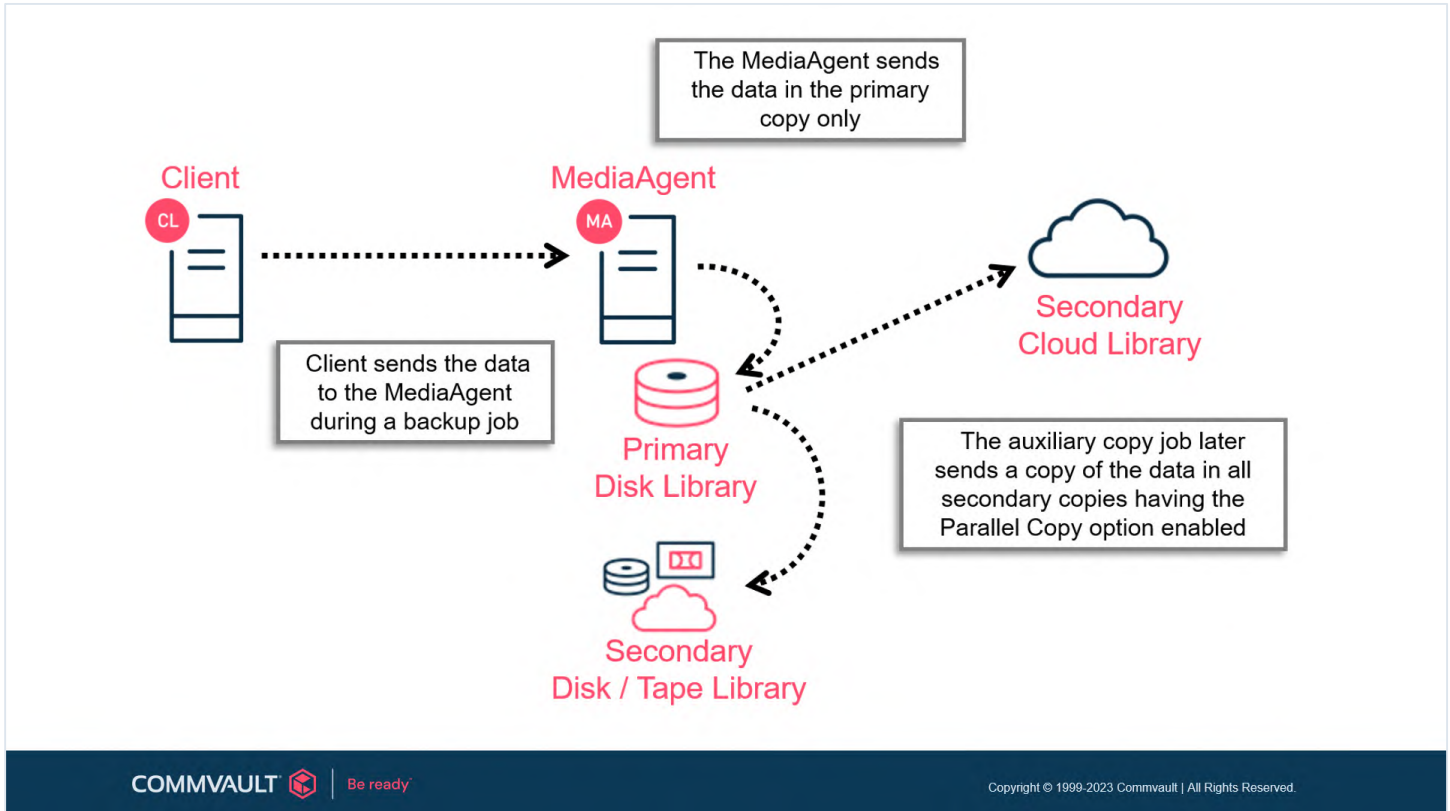
Consider a scenario where the primary library receiving the client streams is a disk library, and two secondary libraries are cloud and tapes. If Inline Copy is enabled on both secondary copies, the three copies are performing at the speed of the slowest target. In this case, let's assume it is the WAN link to reach the cloud library. It might result in tape drive buffers not filling up quickly enough. The tapes therefore must constantly be paused and repositioned, reducing the lifespan of tapes and drives significantly.

Configure Inline Copy



1. Expand the storage policy > right-click the storage policy secondary copy > Properties
2. Check the box for **Enable Inline Copy** to enable inline copy with backups
3. Click **OK** to confirm that you understand both data paths share a common MediaAgent

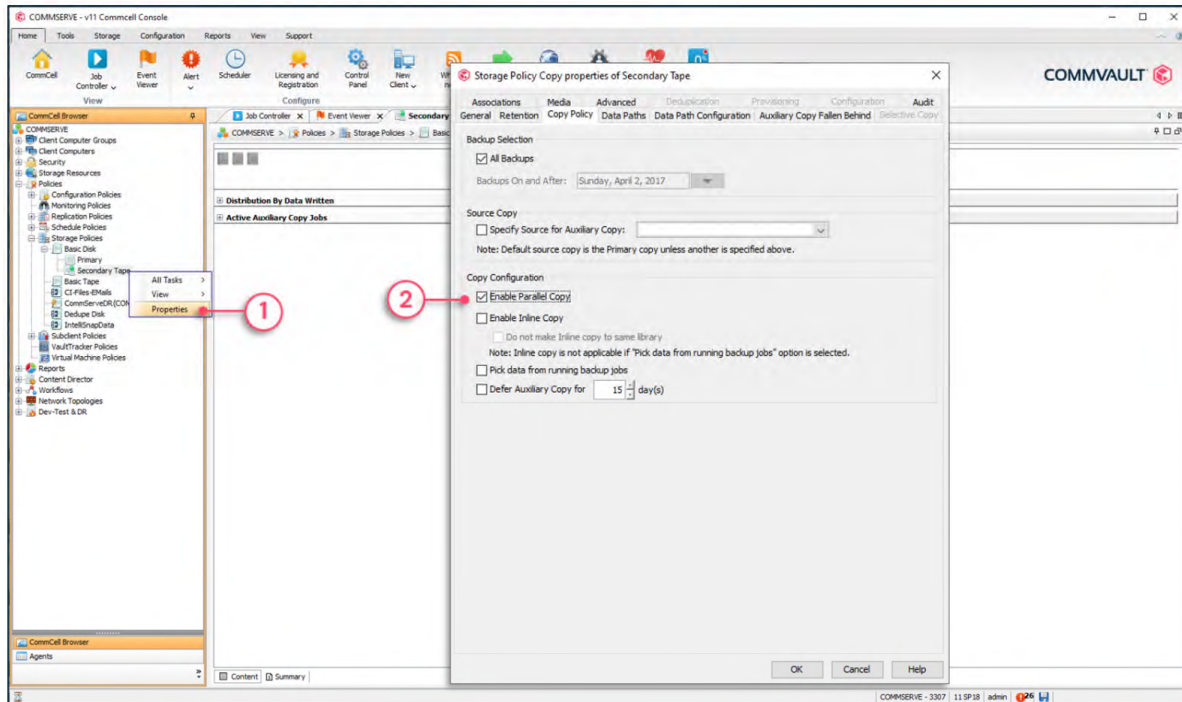
Parallel Copy



A parallel copy generates two secondary copy jobs concurrently when an auxiliary copy job runs. Both secondary copies must have the 'Enable Parallel copy' option selected and the destination libraries must be accessible from the same MediaAgent. Performance is based on the speed of the slowest target.

There is an advantage of using parallel copy rather than inline copy to create multiple secondary copies. A parallel copy is executed by an auxiliary copy schedule and is independent of the backup job, so the backup performance is not negatively impacted as it is for inline copy.

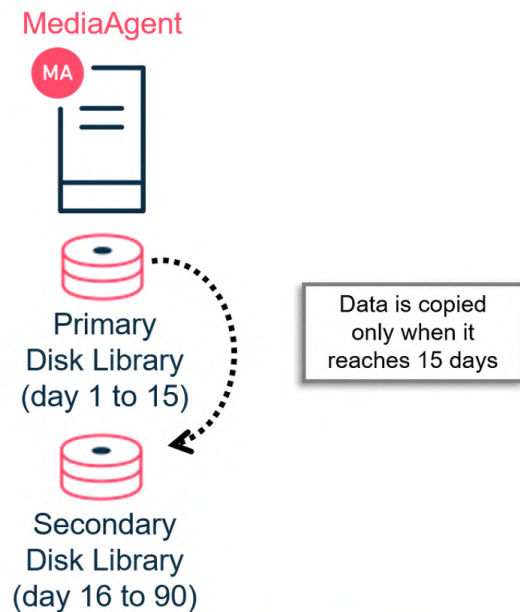
Configure Parallel Copy



1. Expand the storage policy> right-click the storage policy secondary copy > Properties
2. Check the box for **Enable Parallel Copy** to enable parallel copy

Deferred Copy

Storage Policy Copy	Retention	Deferred Copy
Primary	15 days	
Secondary	90 days	15 days



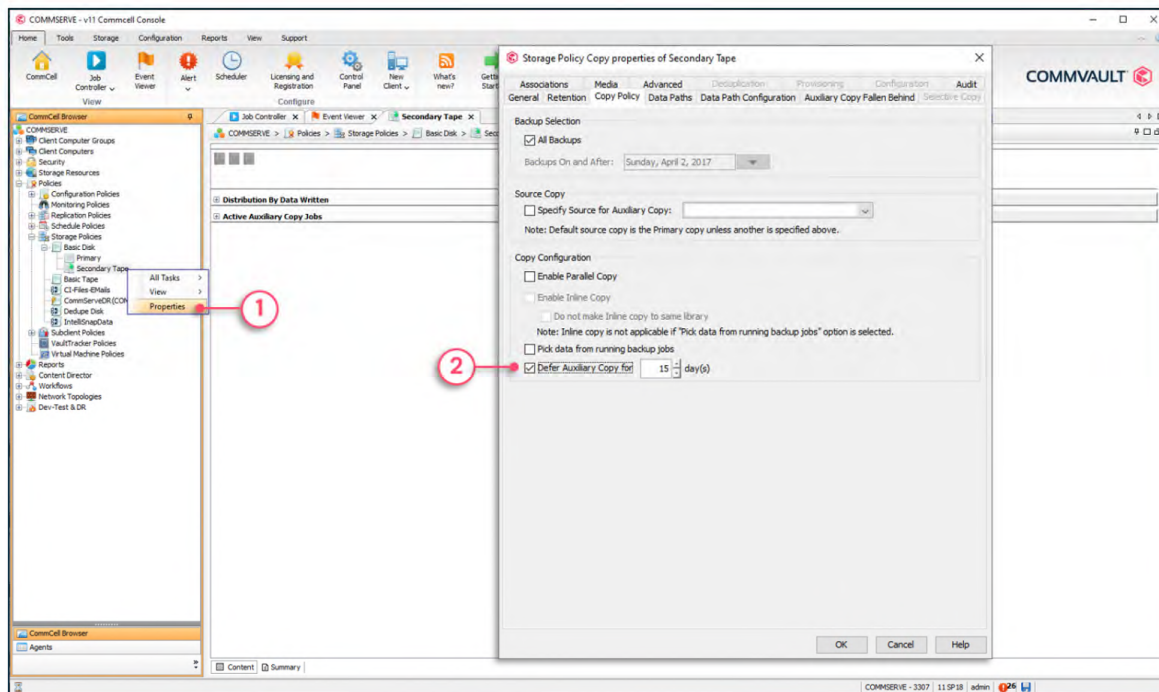
Deferring an auxiliary copy delays a copy from running for a specified number of days. Data is not aged from the source location, regardless of the retention on the source, until the auxiliary copy is completed. This option is traditionally used in Hierarchical Storage Management (HSM) strategies where data will remain in a storage policy copy for a certain period. After that time, the data is copied to another storage policy copy and deleted from the source during the next data aging job.

Commvault recommends copying data to multiple HSM copies to provide for disaster recovery and HSM archiving.

Example

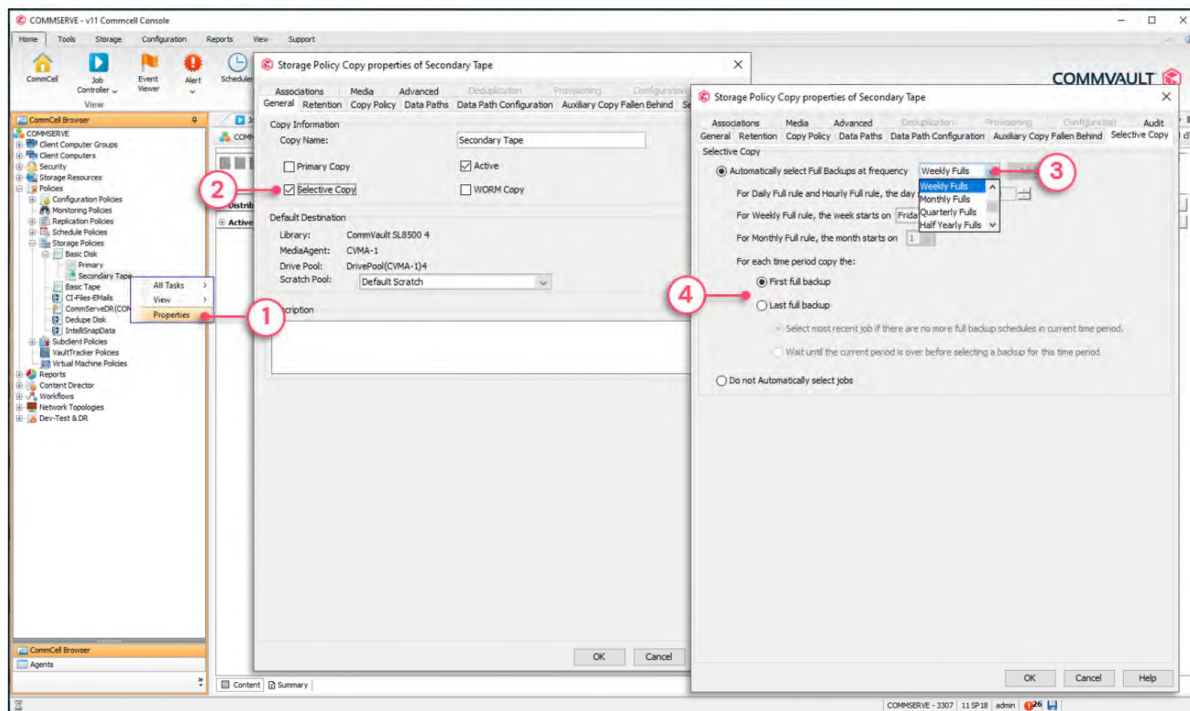
Consider a scenario where a MediaAgent has a costly performant disk library that has reached full capacity. The storage unit is already fully expanded. A larger, cheaper, but lower performant unit is acquired. Instead of just adding mount paths to the actual library (which may not use the performant disks for the most recent data) an HSM strategy leveraging the deferred copy option can be used. The storage policy must have the primary copy stored in the performant library, and a secondary copy in the larger library.

Configure Deferred Copy



1. Expand the storage policy > right-click the storage policy secondary copy > Properties
2. Check the box for **Defer Auxiliary Copy for...** and define how long the copy will be deferred

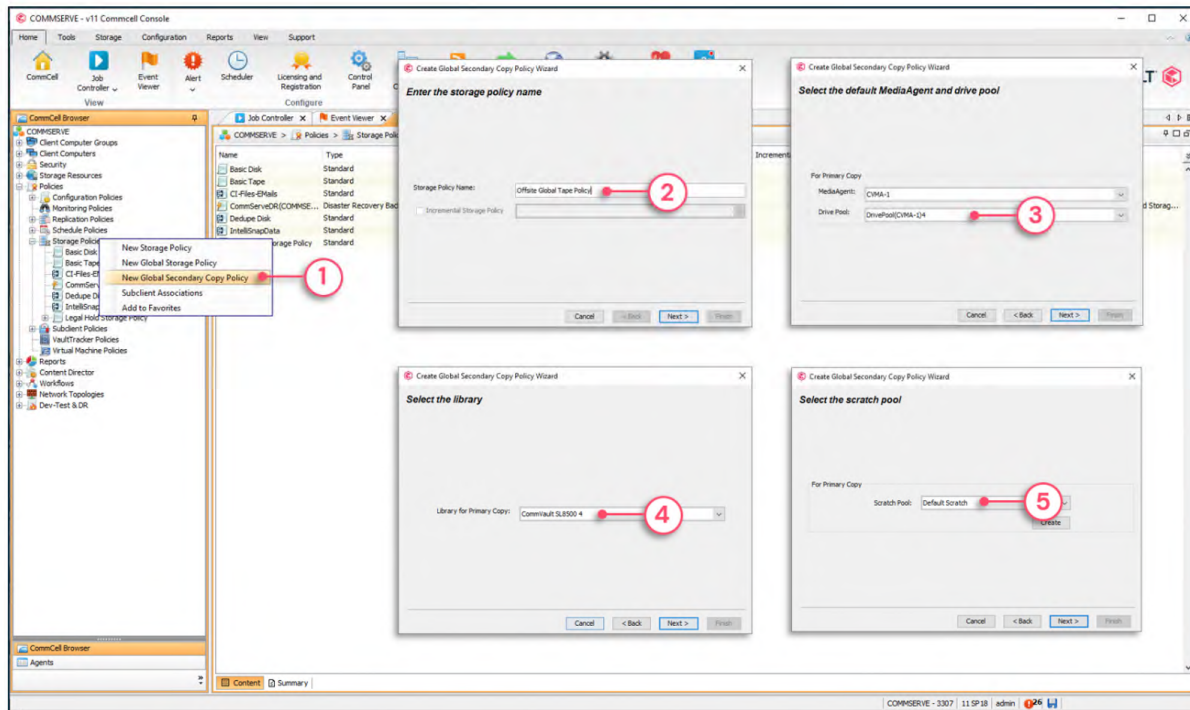
Selective Copy



A selective copy allows automatic selection of specific full backups or manual selection of any backup for additional protection. A selective copy can be scheduled on a specific cadence or frequency (number of cycles). Selective copies are ideal for sending data sent off-site weekly, monthly, quarterly, or yearly.

1. Right-click the **secondary copy** > **Properties**
2. Check the box for **Selective Copy**
3. Select the full backups to copy from the dropdown list
4. Select the first or last full backup of the period

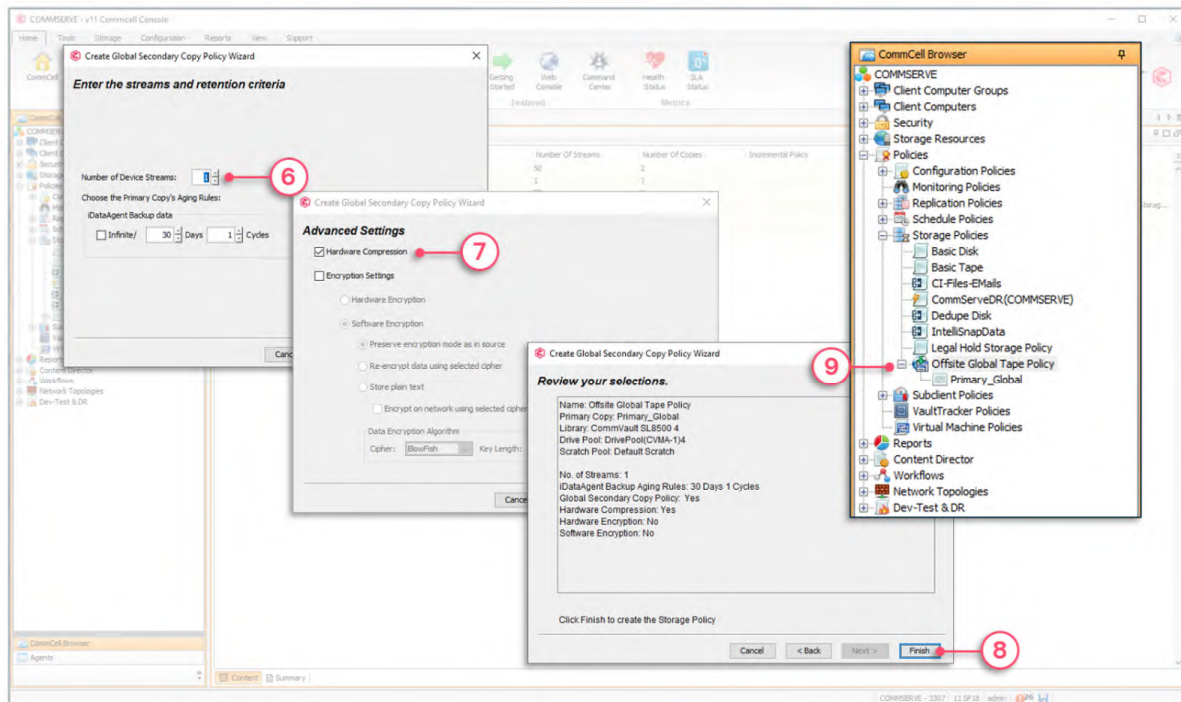
Global Secondary Copy



A global secondary copy policy allows multiple storage policy secondary copies using a tape data path to be associated with a single global secondary copy. This is based on the same concept as global deduplication policies, but global secondary copies only apply to tape copies. If multiple secondary copies require the same retention and encryption settings, using a global secondary copy reduces the number of tapes required during auxiliary copy operations and improves performance.

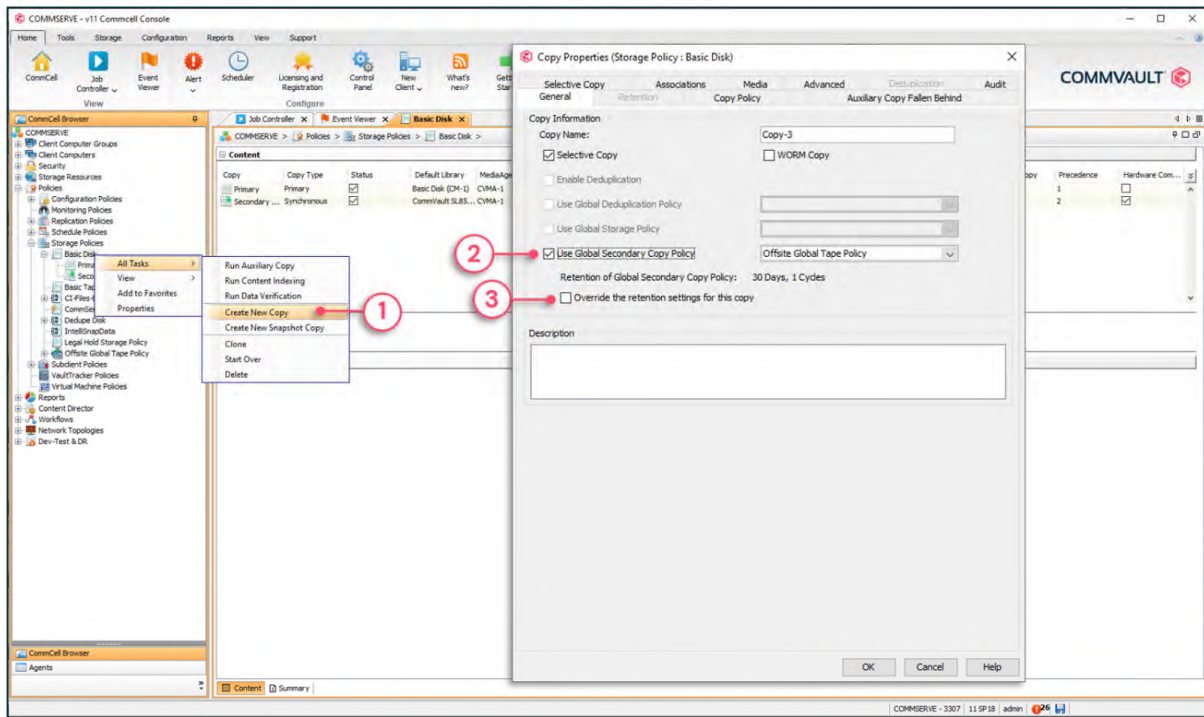
To configure and use a global secondary copy, the global secondary copy policy first needs to be created. Then, in every storage policy that will use it, create a secondary copy associated to the global secondary copy policy.

1. **Right-click Storage Policies > New Global Secondary Copy**
2. Enter a name for the policy
3. Select the tape library
4. Select the MediaAgent and drive pool
5. Select the scratch pool



6. Set the number of streams and the retention for the global copy
7. Enable or disable compression
8. Review settings and click **Finish**
9. The global secondary copy is listed under Storage Policies

Create a Secondary Copy and Assign it to a Global Secondary Copy



1. Right-click storage policies > All Tasks > Create New Copy
2. Check the box for **Use Global Secondary Copy Policy** and select the copy from the dropdown list
3. Choose to use global retention or select **Override** and set the retention

Storage Policies Tasks

Jobs can be managed by viewing the job history for a storage policy copy, including:

- Changing retention for a job
- Deleting a job
- Selecting a job to be included in an auxiliary copy job
- Browsing and finding data to recover
- Selecting a job for data verification

View Jobs in a Storage Policy Copy

1. Click the **Jobs** link in the left sidebar.

2. Check the box for **Include All Client Computers** to include all computers or uncheck the box to select specific clients.

3. Choose the backup type to include.

4. Check the box for **Time Range** to specify a time range, or uncheck the box to select all.

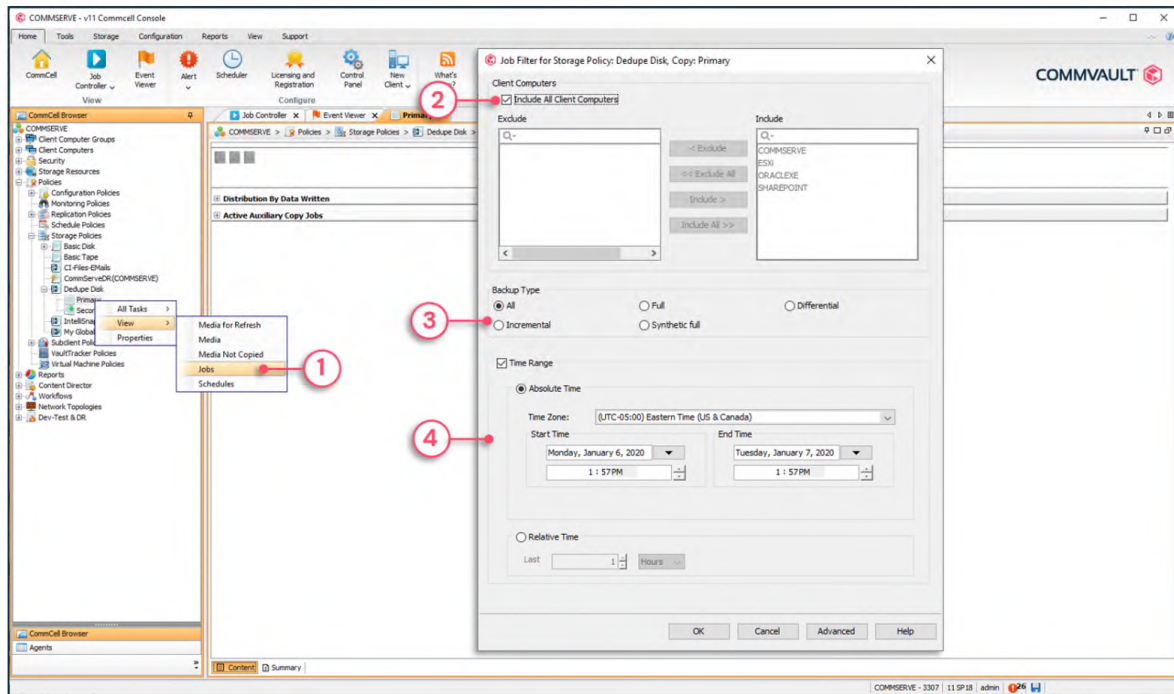
5. All jobs matching the criteria are displayed in the summary view.

Job ID	Status	Client	Agent Type	Instance	Backup Set	Subclient	Job Type	Cycles/S...	Start Time	End Time	Duration	Stub Del...	Size of A...	Data Writ...	Size g	
16345	Compl...	vCenterMain Virtual Se...	VMware	defaultBac...	ProdVM	Incremental	1 / 3	3/27/2017...	3/27/2017...	00:02:39	N/A	1.4 MB	\$19.18 KB	63		
14246	Compl...	vCenterMain Virtual Se...	VMware	defaultBac...	ProdVM	Incremental	1 / 2	3/25/2017...	3/25/2017...	00:03:02	N/A	1.4 MB	\$22.68 KB	63		
13246	Compl...	vCenterMain Virtual Se...	VMware	defaultBac...	ProdVM	Full	1 / 1	3/24/2017...	3/24/2017...	00:09:32	N/A	15.33 GB	1.08 GB	92		
10245	Compl...	ESXi	Virtual Se...	VMware	defaultBac...	TestSubcl...	Full	1 / 1	8/11/2016...	8/11/2016...	00:03:09	N/A	5.31 GB	2.73 GB	48	
8238	Compl...	COMMSER...	Windows ...	defaultBac...	D088Backup	Full	29 / 1	1/31/2016...	1/31/2016...	00:05:44	N/A	13.82 MB	2.8 MB	79		
7234	Compl...	COMMSER...	Windows ...	defaultBac...	D088Backup	Full	28 / 1	1/16/2016...	1/16/2016...	00:00:22	N/A	13.82 MB	2.8 MB	79		
7231	Compl...	COMMSER...	Windows ...	defaultBac...	D088Backup	Full	27 / 1	1/16/2016...	1/16/2016...	00:00:26	N/A	13.82 MB	2.8 MB	79		

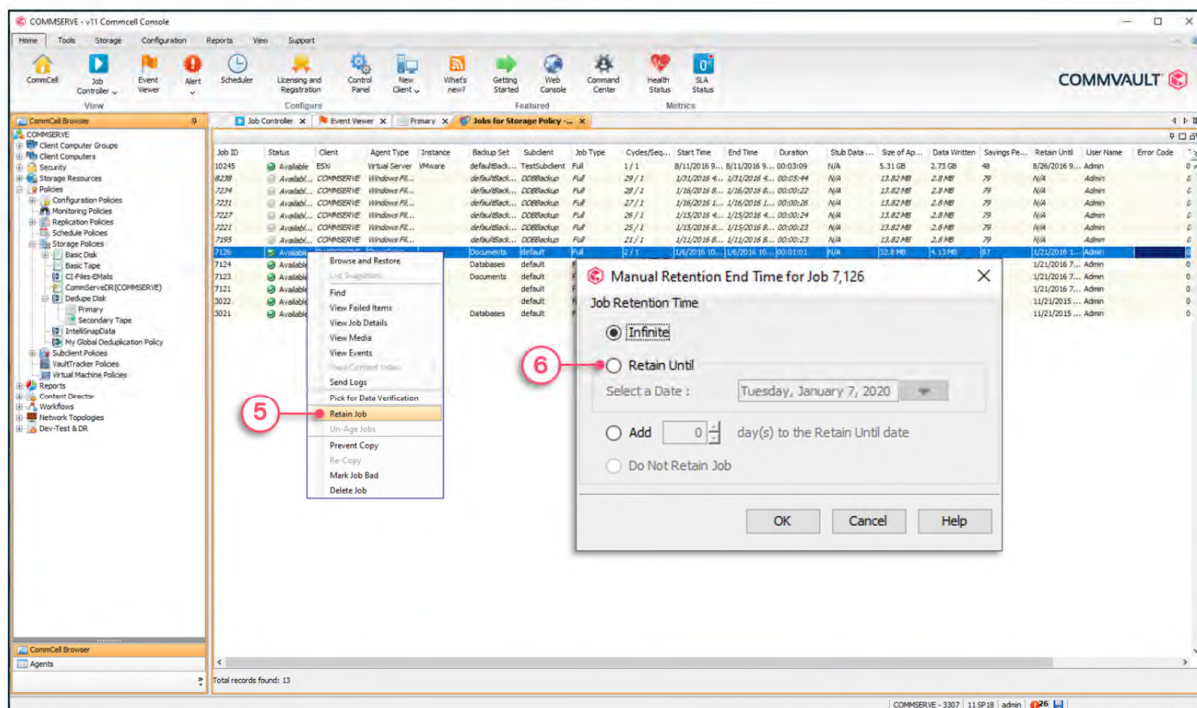
All jobs can be displayed, or if required, displayed for a specific time range and/or filtered out by specific subclients. Right-click any job for additional tasks.

1. Right-click the storage policy copy > View > Jobs
2. Check the box for **Include All Client Computers** to include all computers or uncheck the box to select specific clients
3. Choose the backup type to include
4. Check the box for **Time Range** to specify a time range, or uncheck the box to select all
5. All jobs matching the criteria are displayed in the summary view

Change Job Retention in a Storage Policy Copy



1. Expand the storage policy > right-click the copy > View > Jobs
2. Check the box for **Include All Client Computers** or select specific computers
3. Choose the backup type
4. Specify a time range



5. Right-click the job > Retain Job

6. Set the new retention

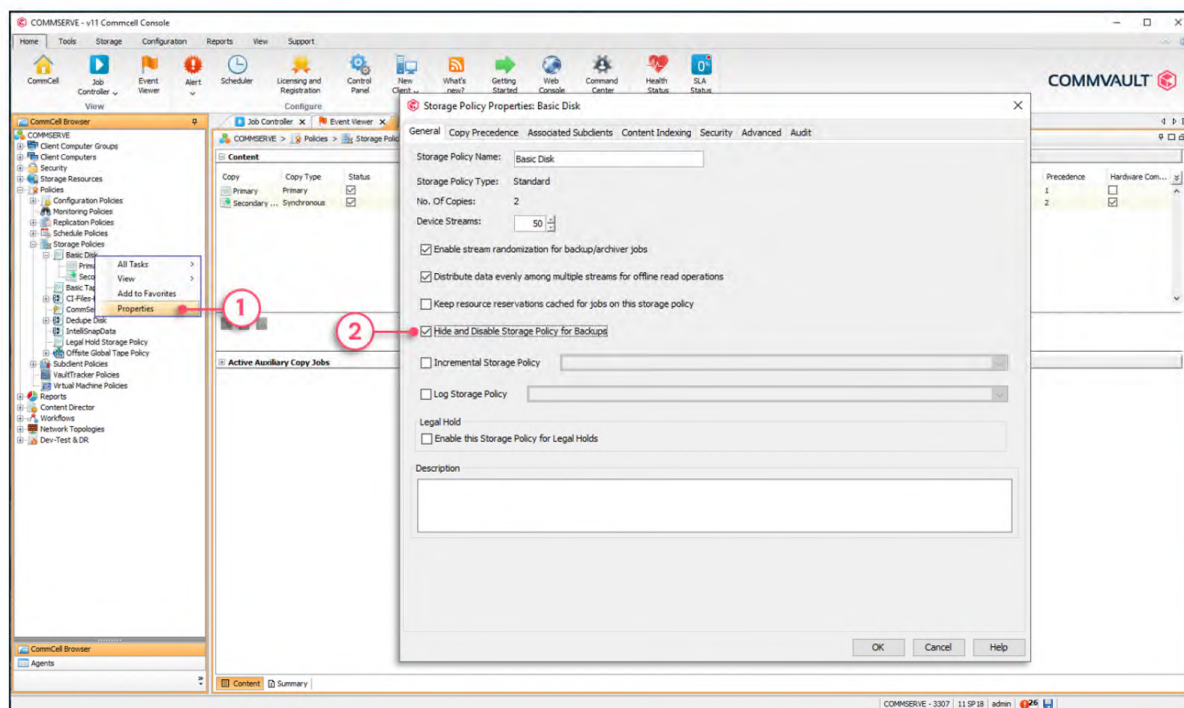
Delete a Job from the Storage Policy Copy

The screenshot shows the Commvault console interface. On the left, the 'CommCell Browser' tree is visible. The main pane displays a table of jobs for a specific storage policy. A right-click context menu is open over job 2021, with the 'Delete Job' option highlighted. A dialog box is displayed in the foreground, asking 'Do you want dependent jobs to be com...' with the 'Yes (Recommended)' option selected. Red circles with numbers 1 and 2 highlight the 'Delete Job' option and the dialog box, respectively.

Job ID	Status	Client	Agent Type	Instance	Backup Set	Subclient	Job Type	Cycles/Seq...	Start Time	End Time	Duration	Stub Data...	Size of Ap...	Data Written	Savings Pe...	Retain Unit	User Name	Error Code
2021	Available	TestSubClient	Full	1 / 1	8/10/2016 8...	11/1/2016 8...	00:00:50	N/A	20.94 MB	1.13 MB	10	1/26/2016 8...	Admin					
10245	Available	TestSubClient	Full	1 / 1	8/10/2016 8...	11/1/2016 8...	00:00:50	N/A	20.94 MB	1.13 MB	10	1/26/2016 8...	Admin					
8228	Available	TestSubClient	Full	20 / 1	1/20/2016 4...	1/20/2016 4...	00:00:44	N/A	13.82 MB	2.8 MB	79	N/A	Admin					
7224	Available	TestSubClient	Full	20 / 1	1/20/2016 8...	1/20/2016 8...	00:00:22	N/A	13.82 MB	2.8 MB	79	N/A	Admin					
7221	Available	TestSubClient	Full	27 / 1	1/24/2016 1...	1/24/2016 1...	00:00:26	N/A	13.82 MB	2.8 MB	79	N/A	Admin					
7227	Available	TestSubClient	Full	26 / 1	1/25/2016 4...	1/25/2016 4...	00:00:24	N/A	13.82 MB	2.8 MB	79	N/A	Admin					
7221	Available	TestSubClient	Full	21 / 1	1/25/2016 8...	1/25/2016 8...	00:00:23	N/A	13.82 MB	2.8 MB	79	N/A	Admin					
7295	Available	TestSubClient	Full	21 / 1	1/21/2016 8...	1/21/2016 8...	00:00:23	N/A	13.82 MB	2.8 MB	79	N/A	Admin					
7126	Available	TestSubClient	Full	2 / 1	1/6/2016 10...	1/6/2016 10...	00:01:01	N/A	32.8 MB	4.13 MB	87	1/21/2016 1...	Admin					
7124	Available	TestSubClient	Full	2 / 1	1/6/2016 10...	1/6/2016 10...	00:01:01	N/A	32.8 MB	4.13 MB	87	1/21/2016 1...	Admin					
7123	Available	TestSubClient	Full	2 / 1	1/6/2016 10...	1/6/2016 10...	00:01:01	N/A	32.8 MB	4.13 MB	87	1/21/2016 1...	Admin					
7121	Available	TestSubClient	Full	2 / 1	1/6/2016 10...	1/6/2016 10...	00:01:01	N/A	32.8 MB	4.13 MB	87	1/21/2016 1...	Admin					
3022	Available	TestSubClient	Full	2 / 1	1/6/2016 10...	1/6/2016 10...	00:01:01	N/A	32.8 MB	4.13 MB	87	1/21/2016 1...	Admin					
3021	Available	TestSubClient	Full	2 / 1	1/6/2016 10...	1/6/2016 10...	00:01:01	N/A	32.8 MB	4.13 MB	87	1/21/2016 1...	Admin					

1. Right-click the job > Delete Job
2. If the job has dependent jobs, choose if dependent jobs should be deleted

Disable Storage Policies for Backups



Disabling a storage policy hides it from view within the CommCell console. Once hidden, it will not appear in the storage policies list and more importantly, subclients cannot be associated with it. To disable a storage policy, no subclients can be associated with the policy.

Disabling is a preferable alternative to deleting. Deleting a policy is a destructive operation that will purge the policy's data which could be required for restore.

1. **Expand Storage Policies > right-click the policy > Properties**
2. Check the box for **Hide and Disable Storage Policy for Backups** to disable the storage policy for backups

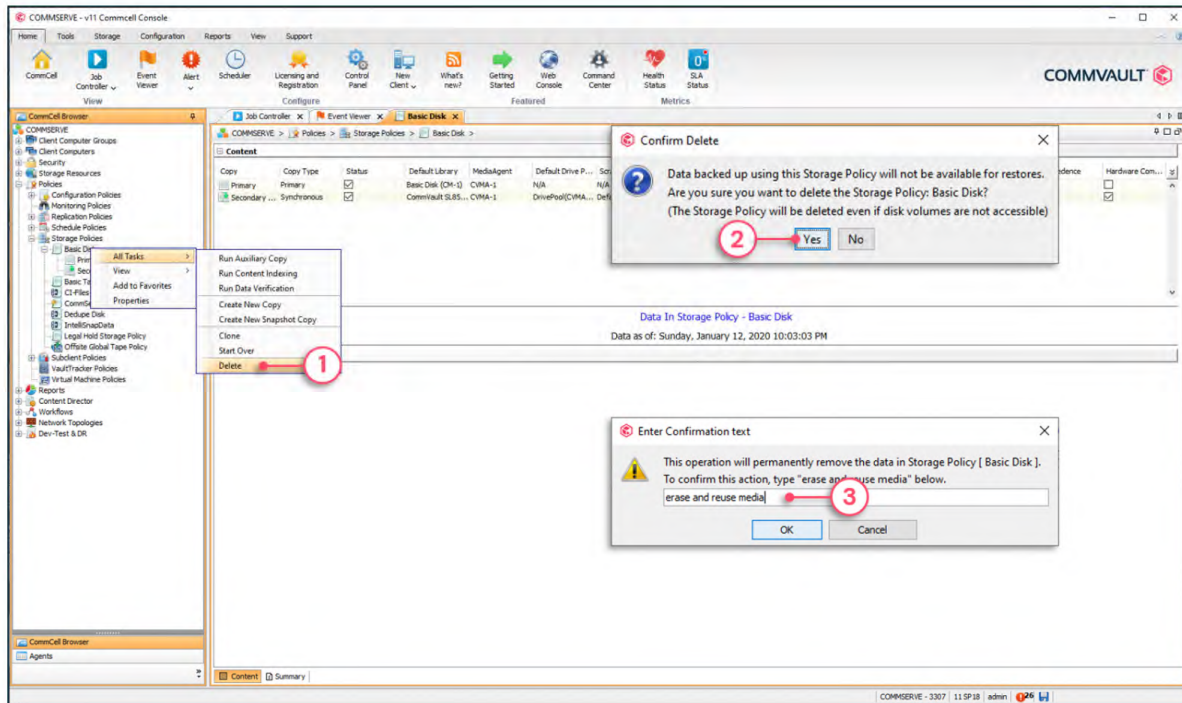
Show Disabled Storage Policies

The screenshot shows the Commvault v11 CommCell Console interface. The 'Storage' tab is selected in the top navigation bar. The 'Media Management' sub-tab is active. The 'Media Management Configuration' dialog box is open, displaying a list of configuration parameters. The parameter 'Show Storage Policies disabled for Backup' is highlighted, and its value is set to 1. A red circle with the number 1 points to the 'Storage' tab, and a red circle with the number 2 points to the value 1 in the configuration dialog.

Parameter	Value
LAN-Free MediaAgent liveliness check interval in Minutes	1440
Library Arm Controller failover interval in Minutes	10
Maximum number of writers for mount path with SAN Data Server configu...	200
Maximum size per batch to process managed disk space	50000
Media Management Maintenance Check Interval in Minutes	15
Mount Path Fragmentation Threshold percentage	100
Number of days in advance to trigger alert for low disk space for disk library	0
Number of volumes for size update	1000
Per-file fragmentation threshold	10
Process volume size updates for cloud mount paths	0
Provide user with MediaAgent management rights additional capabilities f...	1
Redistribute discovered media into spare groups based on barcode patterns	0
Retain scratch media information when deconfiguring library	0
Show hidden libraries	0
Show Mediaagents with no libraries	1
Show Storage Policies disabled for Backup	1
Skip low water mark alerts for mount paths disabled for write.	1
Skip validating Storage Policy Copy master key	0
Split the capacity and free space in a disk library for mount paths pointing...	0
Suppress drive online event for cell-shared library.	1
Suppress media mount failure due to cleaning media while running invent...	0
Suppress repeated wrong media mounted alert for standalone library	0

1. *Storage* tab > **Media Management**
2. Change value for **Show Storage Policies disabled for Backup** to **1**

Delete Storage Policies



If a storage policy is deleted, all protected data associated with the storage policy and all policy copies will be pruned during the next data aging operation. This is a permanent, unreversible action. If there is any chance that you will need to recover the data, consider disabling instead of deleting.

Preparation:

- A storage policy cannot be deleted if subclients are associated with the policy. In the **storage policy properties**, **view the Associations tab** to ensure no subclients are associated with the policy.
- Ensure that there are no jobs being managed by the policy that require to be kept and then exit from the job history. Right click the **storage policy > View > Jobs**. **De-select the option to Specify Time Range then click OK**. This step displays all jobs managed by all copies of the storage policy.

Delete:

1. Expand **Storage Policies > right-click the policy > All Tasks > Delete**
2. Click **Yes** to continue
3. Type the phrase *erase and reuse media* to confirm permanent deletion of data